RTS5264-GR

ONE-LUN SD EXPRESS & UHS-II CARD READER CONTROLLER

DATASHEET

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USING THIS DOCUMENT

This document is intended for the software engineer's reference and provides detailed programming information.

Though every effort has been made to ensure that this document is current and accurate, more information may have become available subsequent to the production of this guide.

REVISION HISTORY

Revision	Release Date	Summary
1.0	2023/12/21	First release

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1. General Description

The RTS5264 is a PCI Express revision 4.0 compliant card reader controller that integrates a PCI Express Transceiver, regulators, memory card access units, and a high speed multiplexer into a single chip. The RTS5264 can support Secure Digital ™, Secure Digital High Capacity ™ (SDHC), Secure Digital eXtended Capacity ™ (SDXC), and Secure Digital Ultra Capacity ™ (SDUC).

The chip fully supports SD UHS-I (Ultra High Speed-I), UHS-II (Ultra High Speed-II), DDR200, DDR225 and SD Express mode. The RTS5264 only requires one PCI Express lane.

The chip integrates 3.3V-to-1.8V and 3.3V-to-1.2V regulators and MOSFET for card power switch, which could dramatically reduce the system BOM cost.

The chip is fully compliant with modern platform requirements for the card reader applications. It supports LTR (Latency Tolerance Reporting), and L1 substate and RTD3 (Runtime D3) functions. It supports Windows modern standby.

The chip is fully compliant with 3.3V/1.8V/1.2V PCIe sideband signal (PERST#, CLKREQ#, WAKE#, PRSNT#).

The chip VID/PID is 10EC/5264.

2. Features

Supports SD Express

Supports 1-Lane 16.0Gbps PCI Express Bus

Supports SDUC up to 128TB

- Compliant with SD Part 1 Physical Layer Specification Version 8.00
- Supports the following memory card interfaces: Secure Digital ™ (SD), SDXC, SDHC, SDUC, Mini-SD, and Micro-SD (T-flash)
- SDUC up to 128TB
- Supports UHS-I SDR-104 (208MHz SD clock), SDR-50 (100MHz SD clock) and DDR-50 (50MHz SD clock), HS and DS.
- Supports DDR200 and DDR225
- Supports SD Express
- Supports UHS-II FD up to FD156 and HD up to HD312
- Supports hardware CRC (Cyclic Redundancy Check) function
- On chip MOSFET/LDO for direct power control of all types memory cards, such as SDVDD1/SDVDD2/SDVDD3
- Built-in variable sampling clock generator to determine correct sampling point for SD UHS-I, DDR200 and DDR225 card.
- Supports Spread Spectrum Clock for SD & UHS-II to reduce EMI effect
- Supports Application Performance Class
- Support SD Command Queue and Cache

Enhance Power Saving

- Supports PCI Express Clock Request (CLKREQ#) to save power consumption
- Supports ASPM L0s/L1 to save power consumption
- Supports Auto-Delink mode to save power consumption
- Supports PCI Express LTR (Latency Tolerance Reporting)
- Supports PCI Express L1 Sub-state (L1.snooze, L1.off)

- Supports PCI Express RTD3 (Runtime D3)
- Supports Modern Standby
- Built-in 3.3V to 1.2V regulator

Supports Latest Platform

- Supports 3.3V, 1.8V, 1.2V PCIe sideband signal (PERST#, CLKREQ#, WAKE#, PRSNT#)
- 32-pin QFN 4x4 package
- Supports eFUSE configuration

3. System Applications

- Notebook PC, All-In-One PC, and other systems with SD card access requirements

4. Block Diagram

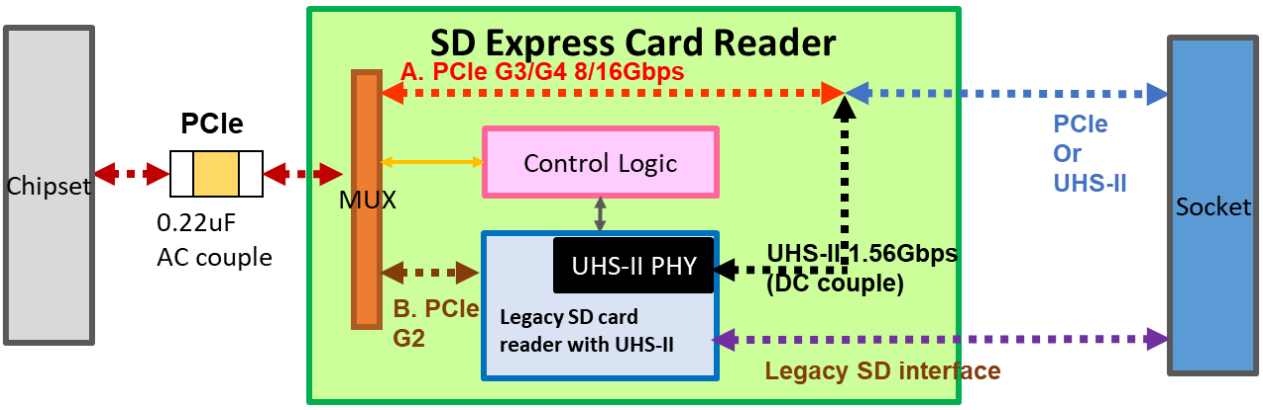


Figure 1. Block Diagram

5. Product Features

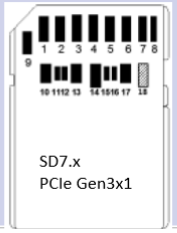
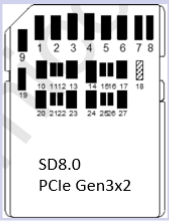
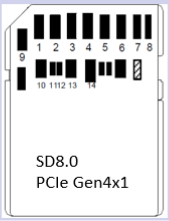
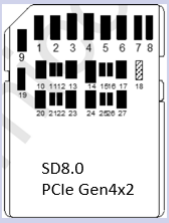
SD Express/SD7.x/SD8.0

SD Express merges PCIe and NVMe interfaces into SD memory cards, together with legacy SD interface. An SD card with PCIe interface is known as an SD Express card.

SD Express was proposed in SD7.x and given enhanced speed in SD8.0.

- SD7.x: SD Express can achieve speeds of up to 985MB/s by PCIe Gen3 1-lane
- SD8.0: SD Express has more speed combinations, and can support PCIe Gen3 2-lane (1969MB/s), PCIe Gen4 1-lane (1969MB/s), or PCIe Gen4 2-lane (3938MB/s)

There is one design for SD7.x, and three different designs for SD8.0. Select the correct option to fit your design.

	SD7.x	SD8.0		
PCIe speed/lane	8Gbps x1	8Gbps x2	16Gbps x1	16Gbps x2
Abbreviation	PCIe G3L1	PCIe G3L2	PCIe G4L1	PCIe G4L1
Max performance	985MB/s	1969MB/s	1969MB/s	3938MB/s
Card Max Power consumption	1.8W	2.8W	2.8W	4W
Micro SD type	Yes	NA	Yes	NA
SDVDD1	Yes	Yes	Yes	Yes
SDVDD1a (NEW)	NA	Yes	Yes	Yes
SDVDD2	Yes	Yes	Yes	Yes
SDVDD2a (NEW)	NA	Yes	NA	Yes
SDVDD3	Option	Option	Option	Option
Standard SD type				

SD Ultra Capacity (SDUC)

The old SDXC spec was limited to 2TB. The SDUC can support more addresses and extends the size to 128TB for legacy SD interface.

PCI Express

The PCIe Gen3 8Gbps l-lane can provide up to 985MB/s bandwidth (Theoretically). The PCIe Gen3 8Gbps 2-lane can provide up to 1969MB/s bandwidth (Theoretically). PCIe Gen4 l-lane can provide up to 1969MB/s bandwidth (Theoretically). PCIe Gen4 2-lane can provide up to 3938MB/s bandwidth (Theoretically).

LTR (Latency Tolerance Report)

LTR is an active message from device to root, which will report device behaviors. Regarding its tolerance of read/write service latencies. That can help system to estimate whole system service routine and enhance efficiency of performance/power saving.

L1 Sub-State

L1 sub-state is a new approach for power management of PCIe interfaces, and is used to enhance power saving effects. It can enter a deeper power saving state than traditional L1, called L1.0.

When the L1.0function enabled, the root and device will use the sideband signal CLKREQ# to negotiate the L1 sub-state entry and exit and allow the device the opportunity to turn off most of the PCIe block.

RTD3

Traditionally the system power state indicated the whole system's components power state. RTD3 is provided to allow the device power state to vary from the system power state. E.g., the device can be in a D3 state when system is in an S0 state.

The system could provide individual power to control each devices state. On a wakeup from RTD3 state, the device needs to connect its Wake# to the system. If the system implements individual power for each device, then AUX-power is required for support D3cold state. However, Wake# and Aux-Power both are required to support RTD3.

Application Performance Class

The Application Performance Class enables higher random performance in an SD card. The SD Association has defined Application Performance Class 1 (A1) and Application Performance Class 2 (A2) standards. An A1 card should meet at least 1500 Input-Output access Per Second (IOPS) in read and 500 IOPS in write.

To further enhance the random performance, Class 2 (A2) introduces new technology such as command queue (CQ) and cache. An A2 SD card should meet 4000 IOPS in read and 2000 IOPS in write.

DDR200/DDR225

DDR200/DDR225 is an enhanced mode of SD card. DDR200/DDR225 is not approved by the SD Association, but some card vendors uses this mode as an 'enhanced mode'. It can provide more bandwidth than UHS-I SDR104, but requires more care in the PCB layout.

6. Pin Assignments

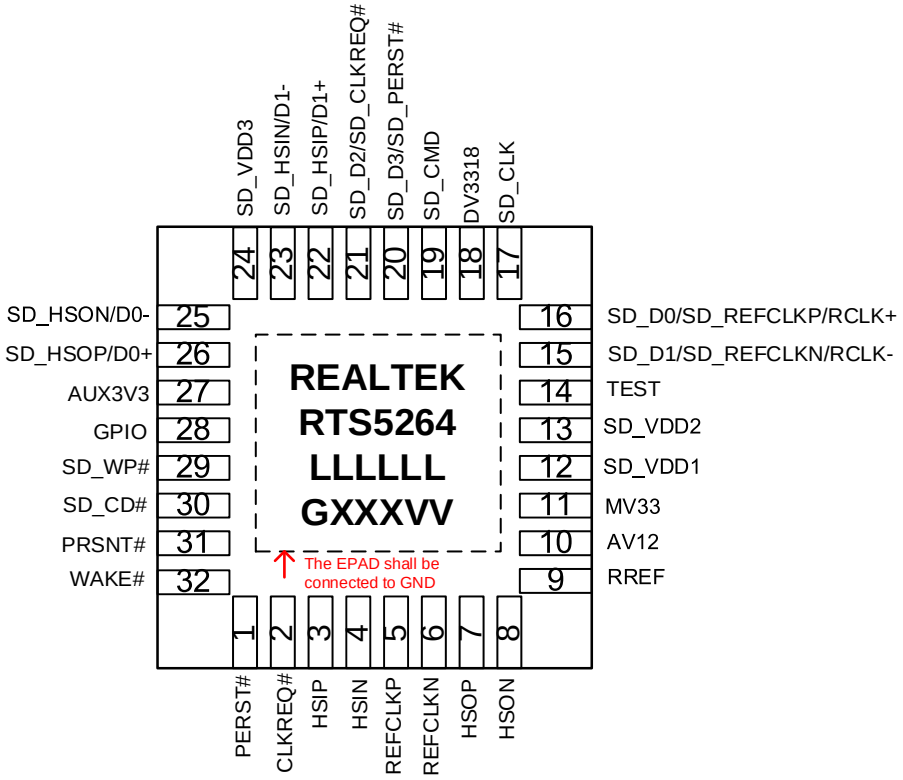


Figure 2. Pin Assignments

6.1. Package Identification

Green package is indicated by the ‘G’ in GXXXVV (Figure 2).

7. Pin Descriptions

7.1. Pin Overview

Table 1. Pin Overview

Pin No.	Symbol	I/O Type	Description
1	PERST#	I	PCI Express Reset from host
2	CLKREQ#	I/O	Clock request signal to PCIe host, and will be used as input direction when enters L1 sub-states
3	HSIP	I	PCIe RX D+ signal
4	HSIN	I	PCIe RX D- signal
5	REFCLKP	I	100MHz reference clock+
6	REFCLKN	I	100MHz reference clock-
7	HSOP	O	PCIe TX D+ signal
8	HSON	O	PCIe TX D- signal
9	RREF	I	Connect external resistor (6.2K ± 1%) to GND
10	AV12	O	1.2V power output
11	MV33	I	3.3V main power input
12	SDVDD1	O	3.3V power for the memory cards
13	SDVDD2	O	1.8V power for the memory cards
14	TEST	I/O	For TEST purposes. Please follow the schematic. If you want to colay with an old chip, reserve a resistor option. Refer to the colay schematic for details
15	SD_D1/SD_REFCLKN/RCLK-	I/O	SD Data 1 (SD_D1),SD Express REFCLK- (SD_REFCLKN) and UHS-II RCLK-
16	SD_D0/SD_REFCLKP/RCLK+	I/O	SD Data 0 (SD_D0),SD Express REFCLK+ (SD_REFCLKP) and UHS-II RCLK+
17	SD_CLK	I/O	SD Clock signal (SD_CLK)
18	DV3318	O	3.3V/1.8V power
19	SD_CMD	I/O	SD CMD signal (SD_CMD)
20	SD_D3/SD_PERST#	I/O	SD Data 3 (SD_D3) and SD Express PERST# (SD_PERST#)
21	SD_D2/SD_CLKREQ#	I/O	SD Data 2 (SD_D2) and SD Express CLKREQ# (SD_CLKREQ#)
22	SD_HSIP/D1+	I/O	SD Express PCIe RX D+ input signal and UHS-II RX D1+ input/output signal
23	SD_HSIN/D1-	I/O	SD Express PCIe RX D- input signal and UHS-II RX D1- input/output signal
24	SDVDD3	O	1.2V power for the memory cards or UHS-II PHY
25	SD_HSON/D0-	I/O	SD Express PCIe TX D- output signal and UHS-II TX D0- output/input signal
26	SD_HSOP/D0+	I/O	SD Express PCIe TX D+ output signal and UHS-II TX D0+ output/input signal
27	AUX3V3	I	3.3V auxiliary power input
28	GPIO	I/O	General purpose input/output. This pin is used for LED application. Pull up for normal usage. Pull up to AUX3V3.
29	SD_WP#	I	SD Write Protect signal
30	SD_CD#	I	SD Card Detection signal

Pin No.	Symbol	I/O Type	Description
31	PRSNT#	I/O	PRSNT# signal for PCIe hot plug. It can notify hot plug events. Pull up for normal usage. Refer to the release schematic for details.
32	WAKE#	I/O	Output direction for WAKE# function
33	E-PAD	GND	The bottom of the package has an E-PAD. The E-PAD shall be connected to the reference ground

7.2. PCI Express Interface

Table 2. PCI Express Interface

Pin No.	Symbol	I/O Type	Description
1	PERST#	I	PCI Express Reset from host
2	CLKREQ#	I/O	Clock request signal to PCIe host and will be used as input direction when enters L1 sub-states
3	HSIP	I	PCIe RX D+ signal
4	HSIN	I	PCIe RX D- signal
5	REFCLKP	I	100MHz reference clock+
6	REFCLKN	I	100MHz reference clock-
7	HSOP	O	PCIe TX D+ signal
8	HSON	O	PCIe TX D- signal
32	WAKE#	I/O	Output direction for WAKE# function

7.3. Card Access Interface

Table 3. Card Access Interface

Pin No.	Symbol	I/O Type	Description
15	SD_D1/SD_REFCLKN/RCLK-	I/O	SD Data 1 (SD_D1),SD Express REFCLK- (SD_REFCLKN) and UHS-II RCLK-
16	SD_D0/SD_REFCLKP/RCLK+	I/O	SD Data 0 (SD_D0),SD Express REFCLK+ (SD_REFCLKP) and UHS-II RCLK+
17	SD_CLK	I/O	SD Clock signal (SD_CLK)
19	SD_CMD	I/O	SD CMD signal (SD_CMD)
20	SD_D3/SD_PERST#	I/O	SD Data 3 (SD_D3) and SD Express PERST# (SD_PERST#)
21	SD_D2/SD_CLKREQ#	I/O	SD Data 2 (SD_D2) and SD Express CLKREQ# (SD_CLKREQ#)
29	SD_WP#	I	SD Write Protect signal
30	SD_CD#	I	SD Card Detection signal

7.4. SD Express Interfaces

Table 4. SD Express Interfaces

Pin No.	Symbol	I/O Type	Description
15	SD_D1/SD_REFCLKN/RCLK-	I/O	SD Data 1 (SD_D1),SD Express REFCLK- (SD_REFCLKN) and UHS-II RCLK-
16	SD_D0/SD_REFCLKP/RCLK+	I/O	SD Data 0 (SD_D0),SD Express REFCLK+ (SD_REFCLKP) and UHS-II RCLK+
20	SD_D3/SD_PERST#	I/O	SD Data 3 (SD_D3) and SD Express PERST# (SD_PERST#)
21	SD2/SD_CLKREQ#	I/O	SD Data 2 (SD_D2) and SD Express CLKREQ# (SD_CLKREQ#)
22	SD_HSIP/D1+	I/O	SD Express PCIe RX D+ input signal and UHS-II RX D1+ input/output signal
23	SD_HSIN/D1-	I/O	SD Express PCIe RX D- input signal and UHS-II RX D1- input/output signal
25	SD_HSON/D0-	I/O	SD Express PCIe TX D- output signal and UHS-II TX D0- output/input signal
26	SD_HSOP/D0+	I/O	SD Express PCIe TX D+ output signal and UHS-II TX D0+ output/input signal

7.5. UHS-II Interfaces

Table 5. UHS-II Interfaces

Pin No.	Symbol	I/O Type	Description
15	SD_D1/SD_REFCLKN/RCLK-	I/O	SD Data 1 (SD_D1),SD Express REFCLK- (SD_REFCLKN) and UHS-II RCLK-
16	SD_D0/SD_REFCLKP/RCLK+	I/O	SD Data 0 (SD_D0),SD Express REFCLK+ (SD_REFCLKP) and UHS-II RCLK+
22	SD_HSIP/D1+	I/O	SD Express PCIe RX D+ input signal and UHS-II RX D1+ input/output signal
23	SD_HSIN/D1-	I/O	SD Express PCIe RX D- input signal and UHS-II RX D1- input/output signal
25	SD_HSON/D0-	I/O	SD Express PCIe TX D- output signal and UHS-II TX D0- output/input signal
26	SD_HSOP/D0+	I/O	SD Express PCIe TX D+ output signal and UHS-II TX D0+ output/input signal

7.6. Power and Ground Interfaces

Table 6. Power and Ground Interfaces

Pin No.	Symbol	I/O Type	Description
10	AV12	O	1.2V power output
11	MV33	I	3.3V main power input
12	SDVDD1	O	3.3V power for the memory cards
13	SDVDD2	O	1.8V power for the memory cards
18	DV3318	O	3.3V/1.8V power
24	SDVDD3	O	1.2V power for the memory cards or UHS-II PHY
27	AUX3V3	I	3.3V auxiliary power input
33	E-PAD	GND	The bottom of the package has an E-PAD. The E-PAD shall be connected to the reference ground

7.7. PAD Status

Table 7. PAD Status

Pin No.	Symbol	No Card Inserted	SD3.0 Card Inserted	SD Express Card Inserted	UHS-II Card Inserted
1	PERST#	No Pull	No Pull	No Pull	No Pull
2	CLKREQ#	No Pull	No Pull	No Pull Bypass mode	No Pull
3	HSIP	PCIe RX Term	PCIe RX Term	No Pull Bypass mode	PCIe RX Term
4	HSIN	PCIe RX Term	PCIe RX Term	No Pull Bypass mode	PCIe RX Term
5	REFCLKP	No Pull	No Pull	No Pull Bypass mode	No Pull
6	REFCLKN	No Pull	No Pull	No Pull Bypass mode	No Pull
7	HSOP	PCIe TX Term	PCIe TX Term	No Pull Bypass mode	PCIe TX Term
8	HSON	PCIe TX Term	PCIe TX Term	No Pull Bypass mode	PCIe TX Term
14	TEST	No Pull	No Pull	No Pull	No Pull
15	SD_D1/SD_REFCLKN/ RCLK-	Pull Down 20	Pull Up 20	No Pull Bypass mode	No Pull Bypass mode
16	SD_D0/SD_REFCLKP/ RCLK+	Pull Down 20	Pull Up 20	No Pull Bypass mode	No Pull Bypass mode
17	SD_CLK	Pull Down 20	Pull Down 20	Pull Down 20	Pull Down 20
19	SD_CMD	Pull Down 20	Pull Up 20	Pull Up 20	Pull Down 20
20	SD_D3/SD_PERST#	Pull Down 20	Pull Up 20	Pull Up 20	Pull Down 20
21	SD_D2/SD_CLKREQ#	Pull Down 20	Pull Up 20	No Pull Bypass mode	Pull Down 20

Pin No.	Symbol	No Card Inserted	SD3.0 Card Inserted	SD Express Card Inserted	UHS-II Card Inserted
22	SD_HSIP/D1+	Pull Down 20	Pull Down 20	No Pull Bypass mode	UHS-II RX Term
23	SD_HSIN/D1-	Pull Down 20	Pull Down 20	No Pull Bypass mode	UHS-II RX Term
25	SD_HSON/D0-	Pull Down 20	Pull Down 20	No Pull Bypass mode	UHS-II TX Term
26	SD_HSOP/D0+	Pull Down 20	Pull Down 20	No Pull Bypass mode	UHS-II TX Term
28	GPIO	No Pull	No Pull	No Pull	No Pull
29	SD_WP#	Pull Down 200	Pull Up 200	Pull Up 200	Pull Up 200
30	SD_CD#	Pull Up 200	Pull Up 200	Pull Up 200	Pull Up 200
31	PRSNT#	No Pull	No Pull	No Pull	No Pull
32	WAKE#	No Pull	No Pull	No Pull	No Pull

Note: Symbol definitions:
Pull Up 20: Internal Pull-Up with 20k ohms. Pull Down 20: Internal Pull-Down with 20k ohms.
Pull Up 200: Internal Pull-UP with 200k ohms.Pull Down 200: Internal Pull-Down with 200k ohms.
TX Term: Internal TX Termination. RX Term: Internal RX Termination.
NoPull: No Pull.
Bypass: Bypass signal.

7.8. Other Interfaces

Table 8. Other Interfaces

Pin No.	Symbol	I/O Type	Description
9	RREF	I	Connect external resistor (6.2K ± 1%) to GND
14	TEST	I/O	For TEST purposes. Please follow the schematic. If you want to colay with an old chip, reserve a resistor option. Refer to the colay schematic for details.
28	GPIO	I/O	General purpose input/output. This pin is used for LED application. Pull up for normal usage. Pull up for AUX3V3.
31	PRSNT#	I/O	PRSNT# signal for PCIe hot plug. It can notify hot plug events. Pull up for normal usage. Refer to the release schematic for details.

8. PCI Configuration Registers

8.1. PCI Configuration Registers Overview

Table 9 is an overview of the PCI configuration registers. For detailed information on the registers go to section 8.2 PCI Configuration Register Descriptions, page 14.

Table 9. PCI Configuration Registers Overview

Offset	Name	Value	Type	Access
00h	Vendor ID	0x10EC	WORD	Read Only
02h	Device ID	0x5264	WORD	Read Only
04h	Command Register	0x0000	WORD	Mixed
06h	Status Register	0x0010	WORD	Mixed
08h	Revision ID	0x01	BYTE	Read Only
09h	Programming Interface	0x00	BYTE	Read Only
0Ah	SubClass	0x00	BYTE	Read Only
0Bh	Class Code	0xFF	BYTE	Read Only
0Ch	Cache Line Size	0x00	BYTE	Read/Write
0Dh	Latency Timer	0x00	BYTE	Read/Write
0Eh	Header Type	0x00	BYTE	Read Only
0Fh	Built-in self test	0x00	BYTE	Mixed
10h	Base Address Register 0	0x00000000	DWORD	Mixed
14h	Base Address Register 1	0x00000000	DWORD	Mixed
18h	Base Address Register 2	0x00000000	DWORD	Mixed
1Ch	Base Address Register 3	0x00000000	DWORD	Mixed
20h	Base Address Register 4	0x00000000	DWORD	Mixed
24h	Base Address Register 5	0x00000000	DWORD	Mixed
28h	Card Bus CIS	0x00000000	DWORD	Read Only
2Ch	Subsystem Vendor ID	0x10EC	WORD	Read Only
2Eh	Subsystem Device ID	0x5264	WORD	Read Only
30h	Expansion ROM	0x00000000	DWORD	Mixed
34h	Capabilities Pointer	0x80	BYTE	Read Only
3Ch	Interrupt Line	0xFF	BYTE	Read/Write
3Dh	Interrupt Pin	0x01	BYTE	Read Only
3Eh	Min_Gnt	0x00	BYTE	Read Only
3Fh	Max_Lat	0x00	BYTE	Read Only
80h	Power Management Interface Capability ID	0x01	BYTE	Read Only
81h	Next Item Pointer	0x90	BYTE	Read Only
82h	PM Capabilities (PMC)	0xF7C3	WORD	Read Only
84h	PM Control/Status(PMCSR)	0x0000	WORD	Mixed
86h	Bridge Support Extensions (PMCSR_BSE)	0x00	BYTE	Read Only
87h	Data	0x00	BYTE	Read Only
90h	MSI Capability ID	0x05	BYTE	Read Only

Offset	Name	Value	Type	Access
91h	Next Item Pointer	0xb0	BYTE	Read Only
92h	MSI Control	0x0080	WORD	Mixed
94h	Message Address	0x00000000	DWORD	Read/Write
98h	Message Upper Address	0x00000000	DWORD	Read/Write
9Ch	MSI Data	0x0000	WORD	Read/Write
A0h	MSI Mask Bits (optional)	0x00000000	DWORD	Read/Write
A4h	MSI Pending Bits (optional)	0x00000000	DWORD	Read Only
B0h	PCI Express Capability ID	0x10	BYTE	Read Only
B1h	Next Item Pointer	0x00	BYTE	Read Only
B2h	PCI Express Capabilities Register	0x0002	WORD	Read Only
B4h	Device Capabilities	0x00008FC2	DWORD	Read Only
B8h	Device Control	0x2010	WORD	Read/Write
BAh	Device Status	0x0010	WORD	Mixed
BCh	Link Capabilities	0x00477C12	DWORD	Read Only
C0h	Link Control	0x0000	WORD	Read/Write
C2h	Link Status	0x1011	WORD	Read Only
D4h	Device Capabilities 2	0x000C0810	DWORD	Read Only
D8h	Device Control 2	0x0000	WORD	Read/Write
DAh	Device Status 2	0x0000	WORD	Read Only
DCh	Link Capabilities 2	0x00000006	DWORD	Read Only
E0h	Link Control 2	0x0002	WORD	Read/Write
E2h	Link Status 2	0x0001	WORD	Read Only
ECh	Epcore register window	0x00000000	DWORD	Read/Write
100h	AER Extended Capability Header	0x14820001	DWORD	Read Only
104h	Uncorrectable Error Status Register	0x00000000	DWORD	Read Only
108h	Uncorrectable Error Mask Register	0x00400000	DWORD	Read/Write
10Ch	Uncorrectable Error Severity Register	0x00462030	DWORD	Read/Write
110h	Correctable Error Status Register	0x00000000	DWORD	Read Only
114h	Correctable Error Mask Register	0x00006000	DWORD	Read/Write
118h	Advanced Error Capabilities and Control Register	0x000000A0	DWORD	Mixed
11Ch	Header Log Registers	0	4DWORD	Read Only
148h	Device Serial Number Extended Capability Header	0x15810003	DWORD	Read Only
14Ch	Serial Number Register (1st DWORD)	0x004CE000	DWORD	Read Only
150h	Serial Number Register (2nd DWORD)	0x00000001	DWORD	Read Only
158h	LTR Extended Capability Header	0x16010018	DWORD	Read Only
15Ch	Max Snoop Latency Register	0x0000	WORD	Read/Write
15Eh	Max No-Snoop Latency Register	0x0000	WORD	Read/Write
160h	L1 Substate Extended Capability Header	0x0001001E	DWORD	Read Only
164h	L1 Substate Capabilities Register	0x00313C1F	DWORD	Read Only
168h	L1 Substate Control 1 Register	0x00000000	DWORD	Read/Write
16Ch	L1 Substate Control 2 Register	0x00000028	DWORD	Read/Write

8.2. PCI Configuration Register Descriptions

8.2.1. Vendor ID Register Offset 0x00

Table 10. Vendor ID Register Offset 0x00

Bits	Description	Attribute	Default
15:0	Vendor identification from PCI SIG.	RO	0x10EC

8.2.2. Device ID Register Offset 0x02

Table 11. Device ID Register Offset 0x02

Bits	Description	Attribute	Default
15:0	Device identification of this product.	RO	0x5264

8.2.3. Command Register Offset 0x04

Table 12. Command Register Offset 0x04

Bits	Description	Attribute	Default
15:11	Reserved.	RsvdP	0
10	Interrupt Disable. 0: Function enabled to generate INTx interrupt messages. 1: Function's ability to generate INTx interrupt messages is disabled.	RW	0
9	Fast back-to-back enable. Does not apply to PCI express and must be 0.	RO	0
8	SERR# Enable. When set, this bit enables the non-fatal and fatal errors detected by the function to be reported to the Root Complex. The function reports such errors to the Root Complex if it is enabled to do so either through this bit or through the PCI express specific bits in the Device Control register. The default value of this bit is 0.	RW	0
7	IDSEL stepping/wait cycle control. Does not apply to PCI express and must be 0.	RO	0
6	Parity error response. In the Status register, the master data parity error bit is set by a requester if its parity error response bit is set and either of the following two conditions occurs: If the requester receives a poisoned completion. If the requester poisons a write request. If the parity error response bit is cleared, the master data parity error status bit is never set.	RW	0
5	VGA palette snoop. Does not apply to PCI express and must be 0.	RO	0
4	Memory write and invalidate. Does not apply to PCI express and must be 0.	RO	0
3	Special cycle enable. Does not apply to PCI express and must be 0.	RO	0

Bits	Description	Attribute	Default
2	Bus master enable. Endpoint 0: Disables an Endpoint function from issuing memory or IO requests. Also disables the ability to generate MSI messages. 1: Enables the Endpoint to issue memory or IO requests, including MSI messages. Root and switch port Not supported by this product	RW	0
1	Memory address space decoder enable. 0: Memory decoder is disabled and Memory transactions targeting this device are not recognized. 1: Memory decoder is enabled and Memory transactions targeting this device are accepted.	RW	0
0	IO address space decoder enable. 0: IO decoder is disabled and IO transactions targeting this device are not recognized. 1: IO decoder is enabled and IO transactions targeting this device are accepted.	RW	0

8.2.4. Status Register Offset 0x06

Table 13. Status Register Offset 0x06

Bits	Description	Attribute	Default
15	Detected parity error.	RW1C	0
14	Signaled system error.	RW1C	0
13	Received master abort.	RW1C	0
12	Received target abort.	RW1C	0
11	Signaled target abort.	RW1C	0
10:9	DEVSEL timing. Only valid for PCI, reserved in PCI express.	RO	0
8	Master data parity error.	RW1C	0
7	Faster back-to-back capable in PCI; reserved in PCI express.	RO	0
6	Previously indicated UDF support. Reserved in 2.2	RO	0
5	66MHz capable. Only Valid for PCI, reserved in PCI express.	RO	0
4	Capability list.	RO	1
3	Interrupt status.	RO	0
2:0	Reserved.	RsvdZ	0

8.2.5. Revision ID Register Offset 0x08

Table 14. Revision ID Register Offset 0x08

Bits	Description	Attribute	Default
7:0	Revision identification.	RO	0x01

8.2.6. Programming Interface Register Offset 0x09

Table 15. Programming Interface Register Offset 0x09

Bits	Description	Attribute	Default
7:0	Programming I/F.	RO	0x00

8.2.7. Sub Class Code Register Offset 0x0A

Table 16. Sub Class Code Register Offset 0x0A

Bits	Description	Attribute	Default
7:0	Sub Class code. Default value of function 0 is 0x00.	RO	0x00

8.2.8. Class Code Register Offset 0x0B

Table 17. Class Code Register Offset 0x0B

Bits	Description	Attribute	Default
7:0	Class code. Default value of function 0 is 0xFF.	RO	0xFF

8.2.9. Cache Line Size Register Offset 0x0C

Table 18. Cache Line Size Register Offset 0x0C

Bits	Description	Attribute	Default
7:0	Cache line size.	RW	0x00

8.2.10. Latency Timer Register Offset 0x0D

Table 19. Latency Timer Register Offset 0x0D

Bits	Description	Attribute	Default
7:0	Not applicable to PCI express and must be hardwired to 0	RW	0x00

8.2.11. Header Type Register Offset 0x0E

Table 20. Header Type Register Offset 0x0E

Bits	Description	Attribute	Default
7	1: Multi-function device 0: Single-function device	RO	0
6:0	Configuration header type. Defines the format of DWORD 4-15 of the function's configuration header. 0: Non-bridge function. 1: PCI-to-PCI bridge (or PCI-X to PCI-X bridge) 2: CardBus bridge	RO	0

8.2.12. BIST Register Offset 0x0F

Table 21. BIST Register Offset 0x0F

Bits	Description	Attribute	Default
7	BIST capable. Should return a one if the function implements a BIST, a zero if it doesn't.	HWInit	0
6	Start BIST. Writing a one into this bit starts the function's BIST. The function resets this bit automatically upon completion. Software should fail the function if the BIST does not complete within two seconds.	RW	0
5:4	Reserved.	RsvdP	0
3:0	Completion code. A value of zero indicates successful completion, while a non-zero result indicates a function-specific error.	RO	0

8.2.13. Base Address Register 0-5 Offset 0x10

Table 22. Base Address Register 0-5 Offset 0x10

Bits	Description	Attribute	Default
31:4	Base address.	RW	0
3	Prefetchable.	RO	0
2:1	Type. 00: 32bit memory address 10: 64bit memory address	RO	0
0	Memory Space Indicator 0: The register is a memory address decoder 1: The register is an IO address decoder	RO	0

8.2.14. CardBus CIS Pointer Register Offset 0x28

Table 23. CardBus CIS Pointer Register Offset 0x28

Bits	Description	Attribute	Default
31:0	CardBus CIS pointer. Not supported by this product.	RO	0

8.2.15. Subsystem Vendor ID Register Offset 0x2C

Table 24. Subsystem Vendor ID Register Offset 0x2C

Bits	Description	Attribute	Default
15:0	Subsystem vendor ID.	RO	0x10EC

8.2.16. Subsystem ID Register Offset 0x2E

Table 25. Subsystem ID Register Offset 0x2E

Bits	Description	Attribute	Default
15:0	Subsystem ID.	RO	0x5264

8.2.17. Expansion ROM Base Address Register Offset 0x30

Table 26. Expansion ROM Base Address Register Offset 0x30

Bits	Description	Attribute	Default
31:11	Expansion ROM bass address.	RO	0
10:1	Reserved	RsvdP	0
0	Expansion ROM Enable.	RW	0

8.2.18. Capability Pointer Register Offset 0x34

Table 27. Capability Pointer Register Offset 0x34

Bits	Description	Attribute	Default
7:2	Points to the first capability register's base address.	RO	'b100000
1:0	Reserved	RsvdP	0

8.2.19. Interrupt Line Register Offset 0x3C

Table 28. Interrupt Line Register Offset 0x3C

Bits	Description	Attribute	Default
7:0	Required if function is capable of generating interrupt requests via an INTx.	RW	0xFF

8.2.20. Interrupt Pin Register Offset 0x3D

Table 29. Interrupt Pin Register Offset 0x3D

Bits	Description	Attribute	Default
7:0	Required if function is capable of generating interrupt requests via an INTx.	RO	0x01

8.2.21. Min_Gnt Register Offset 0x3E

Table 30. Min_Gnt Register Offset 0x3E

Bits	Description	Attribute	Default
7:0	Does not apply to PCI express; must be 0.	RO	0x00

8.2.22. Max_Lat Register Offset 0x3F

Table 31. Max_Lat Register Offset 0x3F

Bits	Description	Attribute	Default
7:0	Does not apply to PCI express; must be 0.	RO	0x00

8.2.23. PM Capability ID Register Offset 0x80

Table 32. PM Capability ID Register Offset 0x80

Bits	Description	Attribute	Default
7:0	PCI express capability ID.	RO	0x01

8.2.24. Next Capability Pointer Register Offset 0x81

Table 33. Next Capability Pointer Register Offset 0x81

Bits	Description	Attribute	Default
7:0	Pointer to next capability register.	RO	0x90

8.2.25. Power Management Capability Register Offset 0x82

Table 34. Power Management Capability Register Offset 0x82

Bits	Description	Attribute	Default
15:11	PME_Support. Bit: Corresponds to PM State D3cold D3hot D2 D1 11: D0	RO	‘b11110
10	D2 Support.	RO	1
9	D1 Support.	RO	1
8:6	Aux_Current. Bit Max Current Required 375mA 320mA 101 270mA 100 220mA 011 160mA 010 100mA 001 55mA 000 0mA	RO	‘b111
5	DSI	RO	0
4	Reserved.	RsvdP	0
3	PME Clock.	RO	0
2:0	Version Field. 001 1.0 010 1.1 011 1.2	RO	‘b011

8.2.26. PM Control/Status Register Offset 0x84

Table 35. PM Control/Status Register Offset 0x84

Bits	Description	Attribute	Default
15	PME_Status.	RW1C	0
14:13	Data_Scale.	RO	0
12:9	Data_Select.	RW	0
8	PME_En.	RW	0
7:4	Reserved.	RsvdP	0
3	No_Soft_Reset.	RO	0
2	Reserved.	RsvdP	0
1:0	PowerState. 00 D0 01 D1 10 D2 11 D3hot	RW	0

8.2.27. Bridge Support Extensions Register Offset 0x86

Table 36. Bridge Support Extensions Register Offset 0x86

Bits	Description	Attribute	Default
7	BPCC_En.	RO	0
6	B2_B3#.	RO	0
5:0	Reserved.	RsvdP	0

8.2.28. Data Offset 0x47

Table 37. Data Offset 0x87

Bits	Description	Attribute	Default
7:0	Data	RO	0x00

8.2.29. MSI Capability ID Offset 0x90

Table 38. MSI Capability ID Offset 0x90

Bits	Description	Attribute	Default
7:0	MSI Capability ID	RO	0x05

8.2.30. MSI Next Item Pointer Offset 0x91

Table 39. MSI Next Item Pointer Offset 0x91

Bits	Description	Attribute	Default
7:0	Next Capability Pointer	RO	0xb0

8.2.31. MSI Control Register Offset 0x92

Table 40. MSI Control Register Offset 0x92

Bits	Description	Attribute	Default
15:9	Reserved	RsvdP	0
8	MSI Per Vector Masking (PVM) supported	RO	0
7	64-bit Address Capable	RO	1
6:4	Multiple Message Enabled. Indicates that multiple Message mode is enabled by system software. The number of Messages enabled must be less than or equal to the Multiple Message Capable value.	RW	0
3:1	Multiple Message Capable	RO	0
0	MSI Enabled When set, INTx must be disabled	RW	0

8.2.32. Message Address Offset 0x94

Table 41. Message Address Offset 0x94

Bits	Description	Attribute	Default
31:2	Lower 32-bit Address	RW	0
1:0	Reserved	RsvdP	0

8.2.33. Message Upper Address Offset 0x98

Table 42. Message Upper Address Offset 0x98

Bits	Description	Attribute	Default
31:0	Upper 32-bit Address. Optional, used only if MSI_64_EN = 1. When MSI_64_EN = 0, then the Upper 32-bit Address register functions as the MSI Data register, and register address 0x0C is not used.	RW	0

8.2.34. MSI Data Offset 0x9C

Table 43. MSI Data Offset 0x9C

Bits	Description	Attribute	Default
15:0	MSI Data. Pattern assigned by system software, bits [4:0] are OR-ed with MSI_VECTOR to generate 32 MSI Messages per function.	RW	0

8.2.35. MSI Mask Bits (Optional) Offset 0xA0

Table 44. MSI Mask Bits (Optional) Offset 0xA0

Bits	Description	Attribute	Default
31:0	MSI Mask. For each Mask bit that is set, the function is prohibited from sending the associated message.	RW	0

8.2.36. MSI Pending Bits (Optional) Offset 0xA4

Table 45. MSI Pending Bits (Optional) Offset 0xA4

Bits	Description	Attribute	Default
31:0	MSI Pending Bits. For each Pending bit that is set, the function has a pending associated message.	RO	0

8.2.37. PCI Express Capability ID Register Offset 0xB0

Table 46. PCI Express Capability ID Register Offset 0xB0

Bits	Description	Attribute	Default
7:0	PCI express capability ID.	RO	0x10

8.2.38. Next Capability Pointer Register Offset 0xB1

Table 47. Next Capability Pointer Register Offset 0xB1

Bits	Description	Attribute	Default
7:0	Pointer to next capability register.	RO	0x00

8.2.39. PCI Express Capability Register Offset 0xB2

Table 48. PCI Express Capability Register Offset 0xB2

Bits	Description	Attribute	Default
15:14	Reserved.	RsvdP	0
13:9	Interrupt message number.	RO	0
8	Slot implemented.	HwInit	0
7:4	Device/Port type.	RO	0
3:0	Capability version.	RO	‘b0010

8.2.40. Device Capability Register Offset 0xB4

Table 49. Device Capability Register Offset 0xB4

Bits	Description	Attribute	Default
31:28	Captured slot power limit scale.	RO	0
27:26	Captured slot power limit value.	RO	0
25:18	Power indicator(Spec 1.0).	RO	0
17:15	Attention indicator(Spec 1.0)	RO	1
14:12	Attention Button(Spec 1.0)	RO	0
11:9	Endpoint L1 acceptable latency. 000: Less than 1µs 001: 1µs to less than 2µs 010: 2µs to less than 4µs 011: 4µs to less than 8µs 100: 8µs to less than 16µs 101: 16µs to less than 32µs 110: 32µs to less than 64µs 111: More than 64µs	RO	‘b111
8:6	Endpoint L0s acceptable latency. 000: Less than 64ns 001: 64ns to less than 128ns 010: 128ns to less than 256ns 011: 256ns to less than 512ns 100: 512ns to less than 1µs 101: 1µs to less than 2µs 110: 2µs to less than 4µs 111: More than 4µs	RO	‘b111
5	Extended tag field supported. 0: 5bit tag field supported 1: 8bit tag field supported	RO	0
4:3	Phantom functions supported.	RO	‘b00
2:0	Max payload size supported. Max data payload size that the function supports for TLP. 000: 128bytes 001: 256bytes 010: 512bytes 011: 1Kbytes 100: 2Kbytes 101: 4Kbytes 110: Reserved 111: Reserved	RO	‘b010

8.2.41. Device Control Register Offset 0xB8

Table 50. Device Control Register Offset 0xB8

Bits	Description	Attribute	Default
15	Bridge Configuration Retry Enable (Bridges).	RW	0
14:12	Max_read_request_size. 000: 128bytes 001: 256bytes 010: 512bytes 011: 1Kbytes 100: 2Kbytes 101: 4Kbytes 110: Reserved 111: Reserved	RW	‘b010
11	Enable no snoop.	RW	0
10	Auxiliary (AUX) power PM enable.	RWS	0
9	Phantom function enable.	RW	0
8	Extended tag field enable.	RW	0
7:5	Max payload size. 000: 128bytes 001: 256bytes 010: 512bytes 011: 1Kbytes 100: 2Kbytes 101: 4Kbytes 110: Reserved 111: Reserved	RW	0
4	Enable relaxed ordering.	RW	1
3	Unsupported request reporting enable.	RW	0
2	Fatal error reporting enable.	RW	0
1	Non-fatal error reporting enable.	RW	0
0	Correctable error reporting enable.	RW	0

8.2.42. Device Status Register 0xBA

Table 51. Device Status Register 0xBA

Bits	Description	Attribute	Default
15:6	Reserved.	RsvdZ	0
5	Transaction pending.	RO	0
4	Aux power detected.	RO	1
3	Unsupported request detected.	RW1C	0
2	Fatal error detected.	RW1C	0
1	Non-fatal error detected.	RW1C	0
0	Correctable error detected.	RW1C	0

8.2.43. Link Capability Register Offset 0xBC

Table 52. Link Capability Register Offset 0xBC

Bits	Description	Attribute	Default
31:24	Port number.	HwInit	0
23	Reserved.	RsvdP	0
22	ASPM Optionality Compliance	HwInit	1
21	Link Bandwidth Notification Capability	RO	0
20	Data Link Layer Link Active Reporting Capable.	RO	0
19	Surprise Down Error Reporting Capable.	RO	0
18	Clock Power Management.	RO	1
17:15	L1 exit latency. 000: Less than 1µs 001: 1µs to less than 2µs 010: 2µs to less than 4µs 011: 4µs to less than 8µs 100: 8µs to less than 16µs 101: 16µs to less than 32µs 110: 32µs to less than 64µs 111: More than 64µs	RO	‘b110
14:12	L0s exit latency. 000: Less than 64ns 001: 64ns to less than 128ns 010: 128ns to less than 256ns 011: 256ns to less than 512ns 100: 512ns to less than 1µs 101: 1µs to less than 2µs 110: 2µs to less than 4µs 111: More than 4µs	RO	‘b111
11:10	Active state power management (ASPM) support.	RO	‘b11
9:4	Max link width. 000000: Reserved. 000001: x1 000010: x2 000100: x4 001000: x8 010000: x16 100000: x32 All other values are reserved.	RO	‘b000001
3:0	Max link speed. 0001b: 2.5Gb/s 0010b: 5.0Gb/s All other values are reserved.	RO	‘b0010

8.2.44. Link Control Register Offset 0xC0

Table 53. Link Control Register Offset 0xC0

Bits	Description	Attribute	Default
15:12	Reserved.	RsvdP	0
11	Link Autonomous Bandwidth Interrupt Enable. When set, this bit enables the generation of an interrupt to indicate that the Link Autonomous Bandwidth Status bit has been set.	RW	0
10	Link Bandwidth Management Interrupt Enable. When set, this bit enables the generation of an interrupt to indicate that the Link Bandwidth Management Status bit has been set.	RW	0
9	Hardware Autonomous Width Disable. Not supported, hardwired to 0.	RW	0
8	Enable Clock Power Management.	RW	0
7	Extended sync.	RW	0
6	Common clock configuration.	RW	0
5	Retrain link.	RW	0
4	Link disable.	RW	0
3	Read Completion Boundary (RCB).	RW	0
2	Reserved.	RsvdP	0
1:0	Active State Power Management (ASPM) control.	RW	0

8.2.45. Link Status Register Offset 0xC2

Table 54. Link Status Register Offset 0xC2

Bits	Description	Attribute	Default
15	Reserved.	RsvdP	0
14	Link Bandwidth Management Status.	RO	0
13	Data Link Layer Link Active (optional).	RO	0
12	Slot clock configuration.	HwInit	1
11	Link training.	RO	0
10	Undefined.	RO	0
9:4	Negotiated link width. 000001: x1 000010: x2 000100: x4 001000: x8 010000: x16 100000: x32 All other values are reserved.	RO	‘b000001
3:0	Link speed. 0001b: 2.5Gb/s All other values are reserved.	RO	‘b0001

8.2.46. Device Capabilities 2 Register Offset 0xD4

Table 55. Device Capabilities 2 Register Offset 0xD4

Bits	Description	Attribute	Default
31:20	Reserved	RsvdP	0
19:18	OBFF Supported. 00b: OBFF Not Supported 01b: OBFF supported using Message signaling only 10b: OBFF supported using WAKE# signaling only 11b: OBFF supported using WAKE# and Message signaling	Hwinit	‘b11
17:14	Reserved	RsvdP	0
13:12	TPH Completer Supported	RO	0
11	LTR Mechanism Supported	RO	1
10	No RO-enabled PR-PR Passing	HWInit	0
9	128-bit CAS Completer Supported	RO	0
8	64-bit AtomicOp Completer Supported	RO	0
7	32-bit AtomicOp Completer Supported	RO	0
6	AtomicOp Routing Supported. Not applicable for EP.	RO	0
5	ARI Forwarding Supported	RO	0
4	Completion Timeout Disable Supported	RO	1
3:0	Completion Timeout Ranges Supported	HWInit	0

8.2.47. Device Control 2 Register Offset 0xD8

Table 56. Device Control 2 Register Offset 0xD8

Bits	Description	Attribute	Default
15	Reserved	RsvdP	0
14:13	OBFF Enable. 00b: Disabled 01b: Enabled using Message signaling [Variation A] 10b: Enabled using Message signaling [Variation B] 11b: Enabled using WAKE# signaling	RW	0
12:11	Reserved	RsvdP	0
10	LTR Mechanism Enable	RW	0
9	IDO Completion Enable	RW	0
8	IDO Request Enable	RW	0
7	AtomicOp Egress Blocking	RW	0
6	AtomicOp Requester Enable	RW	0
5	ARI Forwarding Supported	RW	0
4	Completion Timeout Disable	RW	0
3:0	Completion Timeout Value	RW	0

8.2.48. Device Status 2 Register Offset 0xDA

Table 57. Device Status 2 Register Offset 0xDA

Bits	Description	Attribute	Default
15:0	Reserved	RsvdZ	0

8.2.49. Link Capabilities 2 Register Offset 0xDC

Table 58. Link Capabilities 2 Register Offset 0xDC

Bits	Description	Attribute	Default
31:9	Reserved	RsvdP	0
8	Crosslink Supported	RO	0
7:1	Supported Link Speeds Vector Indicates the supported link speeds of the associated port. For each bit, a value of 1b indicates that the corresponding link speed is supported; otherwise, the link speed is not supported. Bit definitions are: Bit 1: 2.5 GT/s Bit 2: 5.0 GT/s Bit 3: 8.0 GT/s Bits 7:4 reserved	RO	‘b 0000011
0	Reserved	RsvdP	0

8.2.50. Link Control 2 Register Offset 0xE0

Table 59. Link Control 2 Register Offset 0xE0

Bits	Description	Attribute	Default
15:13	Reserved	RsvdP	0
12	Compliance De-emphasis. This bit sets the de-emphasis level in Polling.Compliance state if the entry occurred due to the Enter Compliance bit being 1b. Encodings: 1b: -3.5dB 0b: -6dB When the Link is operating at 2.5 GT/s, the setting of this bit has no effect. Components that support only 2.5 GT/s speed are permitted to hardwire this bit to 0b.	RWS	0
11	Compliance SOS. When set to 1b, the LTSSM is required to send SKP Ordered Sets between sequences when sending the Compliance Pattern or Modified Compliance Pattern Components that support only the 2.5 GT/s speed are permitted to hardwire this field to 0b.	RWS	0
10	Enter Modified Compliance. When this bit is set to 1b, the device transmits Modified Compliance Pattern if the LTSSM enters Polling Compliance substate. Components that support only the 2.5 GT/s speed are permitted to hardwire this bit to 0b.	RWS	0

Bits	Description	Attribute	Default
9:7	Transmit Margin. This field controls the value of the non de-emphasized voltage level at the Transmitter pins. This field is reset to 000b on entry to the LTSSM Polling.Configuration substate. Components that support only the 2.5 GT/s speed are permitted to hardwire this bit to 000b.	RWS	0
6	Selectable De-emphasis. When the Link is operating at 5.0 GT/s speed, this bit is used to control the transmit de-emphasis of the link in specific situations. When the Link is operating at 2.5 GT/s speed, the setting of this bit has no effect. Components that support only the 2.5 GT/s speed are permitted to hardwire this bit to 0b.	HwInit	0
5	Hardware Autonomous Speed Disable. When set, this bit disables hardware from changing the Link speed for device specific reasons other than attempting to correct unreliable Link operation by reducing Link speed. Initial transition to the highest supported common link speed is not blocked by this bit.	RWS	0
4	Enter Compliance. Software is permitted to force a Link to enter Compliance mode at the speed indicated in the Target Link Speed field by setting this bit to 1b in both components on a Link and then initiating a hot reset on the Link. Components that support only the 2.5 GT/s speed are permitted to hardwire this bit to 0b.	RWS	0
3:0	Target Link Speed. For Downstream Ports, this field sets an upper limit on Link operational speed by restricting the values advertised by the Upstream component in its training sequences. Defined encodings are: 0001b: 2.5 GT/s Target Link Speed 0010b: 5.0 GT/s Target Link Speed	RWS	‘b0010

8.2.51. Link Status 2 Register Offset 0xE2

Table 60. Link Status 2 Register Offset 0xE2

Bits	Description	Attribute	Default
15:1	Reserved	RsvdP	0
0	Current De-emphasis Level. When the Link is operating at 5 GT/s speed, this bit reflects the level of de-emphasis. Encodings: 1b: -3.5dB 0b: -6dB The value in this bit is undefined when the Link is operating at 2.5 GT/s speed. Components that support only the 2.5 GT/s speed are permitted to hardwire this field to 0b	RO	1

8.2.52. Epcore Register Window Offset 0xEC

Table 61. Epcore Register Window Offset 0xEC

Bits	Description	Attribute	Default
31:0	Access internal control register window	RW	0

8.2.53. Advanced Error Reporting Extended Capability Header Offset 0x100

Table 62. Advanced Error Reporting Extended Capability Header Offset 0x100

Bits	Description	Attribute	Default
31:20	Next Capability Offset	RO	0x148
19:16	Capability Version	RO	0x2
15:0	PCI Express Extended Capability ID	RO	0x0001

8.2.54. Uncorrectable Error Status Register Offset 0x104

Table 63. Uncorrectable Error Status Register Offset 0x104

Bits	Description	Attribute	Default
31:25	Reserved	RsvdZ	0
24	AtomicOp Egress Blocked Status	RW1CS	0
23	Reserved	RsvdZ	0
22	Uncorrectable Internal Error Status	RW1CS	0
21	Reserved	RsvdZ	0
20	Unsupported Request Error Status	RW1CS	0
19	ECRC Error Status	RW1CS	0
18	Malformed TLP Status	RW1CS	0
17	Receiver Overflow Status	RW1CS	0
16	Unexpected Completion Status	RW1CS	0
15	Completer Abort Status	RW1CS	0
14	Completion Timeout Status	RW1CS	0
13	Flow Control Protocol Error Status	RW1CS	0
12	Poisoned TLP Status	RW1CS	0
11:5	Reserved	RsvdZ	0
4	Data Link Protocol Error Status	RW1CS	0
3:1	Reserved	RsvdZ	0
0	Undefined for PCI Express 1.1 (Was Training Error Status for PCI Express 1.0a)	Undefined	0

8.2.55. Uncorrectable Error Mask Register Offset 0x108

Table 64. Uncorrectable Error Mask Register Offset 0x108

Bits	Description	Attribute	Default
31:25	Reserved	RsvdP	0
24	AtomicOp Egress Blocked Mask	RWS	0
23	Reserved	RsvdP	0
22	Uncorrectable Internal Error Mask	RWS	1
21	Reserved	RsvdP	0
20	Unsupported Request Error Mask	RWS	0
19	ECRC Error Mask	RWS	0
18	Malformed TLP Mask	RWS	0
17	Receiver Overflow Mask	RWS	0
16	Unexpected Completion Mask	RWS	0
15	Completer Abort Mask	RWS	0
14	Completion Timeout Mask	RWS	0
13	Flow Control Protocol Error Mask	RWS	0
12	Poisoned TLP Mask	RWS	0
11:5	Reserved	RsvdP	0
4	Data Link Protocol Error Mask	RWS	0
3:1	Reserved	RsvdP	0
0	Undefined for PCI Express 1.1 (Was Training Error Mask for PCI Express 1.0a)	Undefined	0

8.2.56. Uncorrectable Error Severity Register Offset 0x10C

Table 65. Uncorrectable Error Severity Register Offset 0x10C

Bits	Description	Attribute	Default
31:25	Reserved	RsvdP	0
24	AtomicOp Egress Blocked Severity	RWS	0
23	Reserved	RsvdP	0
22	Uncorrectable Internal Error Severity	RWS	1
21	Reserved	RsvdP	0
20	Unsupported Request Error Severity	RWS	0
19	ECRC Error Severity	RWS	0
18	Malformed TLP Severity	RWS	1
17	Receiver Overflow Severity	RWS	1
16	Unexpected Completion Severity	RWS	0
15	Completer Abort Severity	RWS	0
14	Completion Timeout Severity	RWS	0
13	Completion Timeout Severity	RWS	1
12	Flow Control Protocol Error Severity	RWS	0
11:5	Poisoned TLP Severity	RsvdP	‘b0000001
4	Reserved	RWS	1

Bits	Description	Attribute	Default
3:1	Data Link Protocol Error Severity	RsvdP	0
0	Undefined for PCI Express 1.1 (Was Training Error Severity for PCI Express 1.0a)	Undefined	0

8.2.57. Correctable Error Status Register Offset 0x110

Table 66. Correctable Error Status Register Offset 0x110

Bits	Description	Attribute	Default
31:15	Reserved	RsvdZ	0
14	Corrected Internal Error Status	RW1CS	0
13	Advisory Non-Fatal Error Status	RW1CS	0
12	Replay Timer Timeout Status	RW1CS	0
11:9	Reserved	RsvdZ	0
8	REPLAY_NUM Rollover Status	RW1CS	0
7	Bad DLLP Status	RW1CS	0
6	Bad TLP Status	RW1CS	0
5:1	Reserved	RsvdZ	0
0	Receiver Error Status	RW1CS	0

8.2.58. Correctable Error Mask Register Offset 0x114

Table 67. Correctable Error Mask Register Offset 0x114

Bits	Description	Attribute	Default
31:16	Reserved	RsvdP	0
15	Header Log Overflow Mask	RWS	0
14	Corrected Internal Error Mask	RWS	1
13	Advisory Non-Fatal Error Mask	RWS	1
12	Replay Timer Timeout Mask	RWS	0
11:9	Reserved	RsvdP	0
8	REPLAY_NUM Rollover Mask	RWS	0
7	Bad DLLP Mask	RWS	0
6	Bad TLP Mask	RWS	0
5:1	Reserved	RsvdP	0
0	Receiver Error Mask	RWS	0

8.2.59. Advanced Capabilities and Control Register Offset 0x118

Table 68. Advanced Capabilities and Control Register Offset 0x118

Bits	Description	Attribute	Default
31:9	Reserved	RsvdP	0
8	ECRC Check Enable	RWS	0
7	ECRC Check Capable	RO	1
6	ECRC Generation Enable	RWS	0
5	ECRC Generation Capability	RO	1
4:0	First Error Pointer	ROS	0

8.2.60. Header Log Registers Offset 0x11C

Table 69. Header Log Registers Offset 0x11C

Bits	Description	Attribute	Default
127:96	Header Log Register (fourth DWORD)	-	0
95:64	Header Log Register (third DWORD)	-	0
63:32	Header Log Register (second DWORD)	-	0
31:0	Header Log Register (first DWORD)	-	0

8.2.61. Device Serial Number Extended Capability Header Offset 0x148

Table 70. Device Serial Number Extended Capability Header Offset 0x148

Bits	Description	Attribute	Default
31:20	Next Capability Offset	RO	0x158
19:16	Capability Version	RO	0x1
15:0	PCI Express Extended Capability ID	RO	0x0003

8.2.62. Serial Number Register Offset 0x14C

Table 71. Serial Number Register Offset 0x14C

Bits	Description	Attribute	Default
63:32	Serial Number register (2nd DWORD)	RO	0x0000 0001
31:0	Serial Number register (1st DWORD)	RO	0x00 4CE000

8.2.63. LTR Extended Capability Header Offset 0x158

Table 72. LTR Extended Capability Header Offset 0x158

Bits	Description	Attribute	Default
31:20	Next Capability Offset	RO	0x160
19:16	Capability Version	RO	0x1
15:0	PCI Express Extended Capability ID	RO	0x0018

8.2.64. LTR Max Snoop Latency Register Offset 0x15C

Table 73. LTR Max Snoop Latency Register Offset 0x15C

Bits	Description	Attribute	Default
15:13	Reserved	RsvdP	0
12:10	Max Snoop Latency Scale	RW	0
9:0	Max Snoop Latency Value	RW	0

8.2.65. LTR Max No-Snoop Latency Register Offset 0x15E

Table 74. LTR Max No-Snoop Latency Register Offset 0x15E

Bits	Description	Attribute	Default
15:13	Reserved	RsvdP	0
12:10	Max No-Snoop Latency Scale	RW	0
9:0	Max No-Snoop Latency Value	RW	0

8.2.66. L1 PM Substates Extended Capability Header Offset 0x160

Table 75. L1 PM Substates Extended Capability Header Offset 0x160

Bits	Description	Attribute	Default
31:20	Next Capability Offset	RO	0
19:16	Capability Version	RO	0x1
15:0	PCI Express Extended Capability ID	RO	0x001E

8.2.67. L1 PM Substates Capabilities Register Offset 0x164

Table 76. L1 PM Substates Capabilities Register Offset 0x164

Bits	Description	Attribute	Default
31:24	Reserved	RsvdP	0
23:19	Port T_POWER_ON Value. Along with the Port T_POWER_ON Scale field in the L1 PM Substates Capabilities register sets the time (in µs) that this Port requires the port on the opposite side of Link to wait in L1.2.Exit after sampling CLKREQ# asserted before actively driving the interface. The value of Port T_POWER_ON is calculated by multiplying the value in this field by the scale value in the Port T_POWER_ON Scale field in the L1 PM Substates Capabilities register. Default value is 00101b Required for all	RO	’b00110
18	Reserved	RsvdP	0
17:16	Port T_POWER_ON Scale. Specifies the scale used for the Port T_POWER_ON Value field in the L1 PM Substates Capabilities register. Range of Values: 00b: 2µs 01b: 10µs 10b: 100µs 11b: Reserved Required for all Ports that support L1.2, otherwise this field is of type RsvdP. Default value is 00b	RO	‘b01
15:8	Port Common_Mode_Restore_Time. Time (in µs) required for this Port to re-establish common mode. Required for all Ports that support L1.2, otherwise this field is of type RsvdP.	RO	0x3C
7:5	Reserved	RsvdP	0
4	L1 PM Substates Supported. When set this bit indicates that this Port supports L1 PM Substates. For compatibility with possible future extensions, software must not enable L1 PM Substates unless this bit is Set. Required for both Upstream and Downstream Ports.	RO	1
3	ASPM L1.1 Supported. When set this bit indicates that ASPM L1.1 is supported. Required for both Upstream and Downstream Ports.	RO	1
2	ASPM L1.2 Supported. When set this bit indicates that ASPM L1.2 is supported. Required for both Upstream and Downstream Ports.	RO	1
1	PCI-PM L1.1 Supported. When set this bit indicates that PCI-PM L1.1 is supported, and must be set by all Ports implementing L1 PM Substates. Required for both Upstream and Downstream Ports.	RO	1
0	PCI-PM L1.2 Supported. When set this bit indicates that PCI-PM L1.2 is supported. Required for both Upstream and Downstream Ports.	RO	1

8.2.68. L1 PM Substates Control 1 Register Offset 0x168

Table 77. L1 PM Substates Control 1 Register Offset 0x168

Bits	Description	Attribute	Default
31:29	LTR_L1.2_THRESHOLD_Scale. This field provides a scale for the value contained within the LTR_L1.2_THRESHOLD_Value. Encoding is the same as the LatencyScale fields in the LTR Message (see Section 6.18). The default value for this field is 000b. Hardware operation is undefined if software writes a Not-Permitted value to this field. Required for all Ports that support ASPM L1.2, otherwise this field is of type RsvdP.	RW	0
28:26	Reserved	RsvdP	0
25:16	LTR_L1.2_THRESHOLD_Value. Along with the LTR_L1.2_THRESHOLD_Scale, this field indicates the LTR threshold used to determine if entry into L1 results in L1.1 (if enabled) or L1.2 (if enabled). The default value for this field is 00 0000 0000b. Required for all Ports that support ASPM L1.2, otherwise this field is of type RsvdP.	RW	0
15:8	Common_Mode_Restore_Time. Sets value of TCOMMONMODE (in µs), which must be used by the Downstream Port for timing the re-establishment of common mode, as described in Table 5-xx. Required for Downstream Ports that support L1.2, otherwise this field is of type RsvdP. This field is of type RsvdP for Upstream Ports. Default value is implementation specific.	RW	0
7:4	Reserved	RsvdP	0
3	ASPM L1.1 Enabled. When set this bit enables ASPM L1.1. Required for both Upstream and Downstream Ports. For Ports that do not support ASPM L1.1 this bit is permitted to be hardwired to 0. Default value is 0b.	RW	0
2	ASPM L1.2 Enabled. When set this bit enables ASPM L1.2. Required for both Upstream and Downstream Ports. For Ports that do not support ASPM L1.2 this bit is permitted to be hardwired to 0. Default value is 0b.	RW	0
1	PCI-PM L1.1 Enabled. When set this bit enables PCI-PM L1.1. Required for both Upstream and Downstream Ports. Default value is 0b.	RW	0
0	PCI-PM L1.2 Enabled. When set this bit enables PCI-PM L1.2. Required for both Upstream and Downstream Ports. For Ports that do not support PCI-PM L1.2 this bit is permitted to be hardwired to 0. Default value is 0b.	RW	0

8.2.69. L1 PM Substates Control 2 Register Offset 0x16C

Table 78. L1 PM Substates Control 2 Register Offset 0x16C

Bits	Description	Attribute	Default
31:8	Reserved	RsvdP	-
7:3	T_POWER_ON Value. Along with the T_POWER_ON Scale sets the minimum amount of time (in µs) that the Port must wait in L1.2. Exit after sampling CLKREQ# asserted before actively driving the interface. T_POWER_ON is calculated by multiplying the value in this field by the value in the T_POWER_ON Scale field. Default value is 00101b Required for all Ports that support L1.2, otherwise this field is of type RsvdP.	RW	‘b00101
2	Reserved	RsvdP	0
1:0	T_POWER_ON Scale. Specifies the scale used for T_POWER_ON Value. Range of Values 00b: 2µs 01b: 10µs 10b: 100µs 11b: Reserved Required for all Ports that support L1.2, otherwise this field is of type RsvdP. Default value is 00b	RW	0

9. Notes for Implementation

9.1. PCIe Hot Plug

In order to use this chip, system support for PCIe hot plug is necessary. The SoC or PCH has to support PCIe hot plug. BIOS also has to enable PCIe hot plug related option. As different platforms have different settings, please contact your SoC or PCH/BIOS vendor for detail.

9.2. PRSNT#

In this chip, the PRSNT# pin is used for hot plug notification. You must connect this pin to the SoC, PCH, or EC GPIO. The GPIO connections depend on your system design. Please contact your BIOS or EC engineer for details.

We strongly suggest to configure your SoC, PCH, or EC GPIO to input mode. We suggest to use the same power source as AUX3V3 to pull up PRSNT#.

When the chip drives PRSNT# low, it indicates that a new device has been plugged in. When PRSNT# goes high, it means a device was unplugged.

We recommend to implement PRSNT# for hot plug detection.

Table 79. PRSNT#

PRSNT#	Description
1	Device was not presented
0	Device was presented

9.3. PCIe 1.8V/1.2V Sideband IO

In SD Express mode, the SD Express card acts as a standalone PCIe device. As SD Express cards come from various vendors, we cannot ensure all SD Express cards support 1.8V/1.2V PCIe sideband IO. For the best compatibility, please follow the instructions below. We provide 3 options for customer reference.

1. Add level shift for all PCIe sideband (PERST#, CLKREQ#, WAKE#).

2. Add level shift for CLKREQ# only.

3. If BOM cost is very important for your project, you can disable the CLKREQ# feature for SD Express cards. Refer to BIOS guide 0x814 bit 1 for detail. However, if the SD Express card supports L1 substate, it may result in the SD Express card failing to enter L1 substate, so the system power consumption might be higher when using the SD Express card.

In legacy SD mode, this chip can fully support PCIe 3.3V/1.8V/1.2V sideband IO.

We suggest that the pull up power source for 1.8V or 1.2V PCIe sideband IO should follow the AUX3V3 power sequence.

For more information on using PCIe 1.8V/1.2V sideband IO, contact Realtek FAE.

9.4. SD7.x or SD8.0 design

SD7.x and SD8.0 follow different hardware PCB designs and SoC/PCH/CPU settings, as well as different SD sockets.

SD7.x design is a PCIe Gen3 1-lane design. There is only one option in the SD7.x design. The SD8.0 design is PCIe Gen3 2-lane, PCIe Gen4 1-lane, or PCIe Gen4 2-lane. There are three options in SD8.0 designs. Please select the correct option to fit your design.

9.5. UHS-II Polarity Cannot be Reversed

The polarity of SD_HSIP and SD_HSIN cannot be reversed. The polarity of SD_HSOP and SD_HSON also cannot be reversed.

9.6. Micro SD design

Note that micro SD Express card and micro SD UHS-II card pin locations are different. Some socket vendors provide a ‘combo’ socket. They claim that their socket can support both micro SD Express and micro SD UHS-II card in one micro SD socket. The socket vendor should ensure their combo socket quality.

Also note that micro SD does not have SD_WP#. It should connect to GND for a normal type socket. Refer to the application for more details.

10. Electrical Characteristics

10.1. Absolute Maximum Ratings

Table 80. Absolute Maximum Ratings

Parameter	Rating
Supply Voltage	–0.5V to +3.6V
Operating Ambient Temperature Range	0°C to +70°C
Operating humidity	<90%
Storage Temperature Range With ANTI-STATIC SHIELDING BAG	-40°C to +125°C
Non-operating humidity	<90%
Latchup Current	250mA
Electrostatic Discharge Voltage (HBM)	4000V
Electrostatic Discharge Voltage (MM)	200V
Electrostatic Discharge Voltage (CDM)	500V

10.2. DC Characteristics

10.2.1. System Power and I/O Pin Characteristics

Table 81. System Power and I/O Pin Characteristics

Symbol	Description	Min.	Typ.	Max.	Unit
VDD	Supply Voltage of MV33 and AUX3V3	3.0	3.3	3.6	V
IDD-MV33	Supply Current of MV33 (Note 2)	-	1.5	-	A
IDD-AUX3V3	Supply Current of AUX3V3 (Note 2)	-	375	-	mA
VIH	Input Voltage High	2	-	VDD	V
VIL	Input Voltage Low	-0.3	-	0.8	V
VOH	Output Voltage High	85%* V_{DD}	-	-	V
VOL	Output Voltage Low	-	-	0.4	V
Cin	Input Pin Capacitance	-	-	10	pF
T_P	MV33 and AUX3V3 rising timing	0.5	-	15	ms
T_F	MV33 and AUX3V3 falling timing	-	-	100	ms

Note 1: The TP and TF are the level from 10% ~ 90% of VDD

Note 2: The system power rail capacity of MV33 should reserve at least 1.5A; AUX3V3 is 375mA.

10.2.2. PCIe Sideband 3.3/1.8/1.2V Tolerance I/O Pin Characteristics

Table 82. PCIe sideband I/O Pin Characteristics

Symbol	Description	Min.	Typ.	Max.	Unit
VIH	Input Voltage High	0.78	-	VDD	V
VIL	Input Voltage Low	-0.3	-	0.42	V
Cin	Input Pin Capacitance	-	-	10	pF

Note: PCIe sideband open drain I/O pad can operate at 3.3V, 1.8V and 1.2V input voltage.

10.2.3. Internal Power Output Characteristics

Table 83. Internal Power Output Characteristics

Symbol	Description	Min.	Typ.	Max.	Unit
V_REFF	Bandgap Voltage	-	1.2	-	V
V_AV12	Core Power LDO	-	1.2	-	V
V_SDVDD1	SDVDD1 Output Voltage	-	3.3	-	V
I_SDVDD1	SDVDD1 Supply Current	-	900	-	mA
OCP_SDVDD1	Over Current Protect for SD/SD Express card		1100		mA
V_SDVDD2	SDVDD2 Output Voltage	-	1.8	-	V
I_SDVDD2	SDVDD2 Supply Current	-	600	-	mA
OCP_SDVDD2	Over Current Protect for SD Express card		750		mA
V_SDVDD3	SDVDD3 Output Voltage	-	1.2	-	V
I_SDVDD3	SDVDD3 Supply Current	-	600	-	mA
OCP_SDVDD3	Over Current Protect for SD Express card		750		mA
V_DV3318_SD20	DV3318 LDO for non-UHS-I/UHS-II	-	3.3	-	V
V_DV3318_SD30	DV3318 LDO for UHS-I	-	1.8	-	V
V_DV3318_SDEx	DV3318 LDO for SD Express	-	3.3	-	V

10.2.4. Internal Pull Resister Characteristics

Table 84. Internal Pull Resister Characteristics

Symbol	Description	Min.	Typ.	Max.	Unit
Pull Up 20/PU20	Internal Pull-Up with 20k ohms	10	20	40	K Ohms
Pull Down 20/PD20	Internal Pull-Down with 20k ohms	10	20	40	K Ohms
Pull Up 200/PU200	Internal Pull-Up with 200k ohms	150	200	250	K Ohms
Pull Down 200/PD200	Internal Pull-Down with 200k ohms	150	200	250	K Ohms

10.3. AC Characteristics

10.3.1. Chip Power Sequence Timing

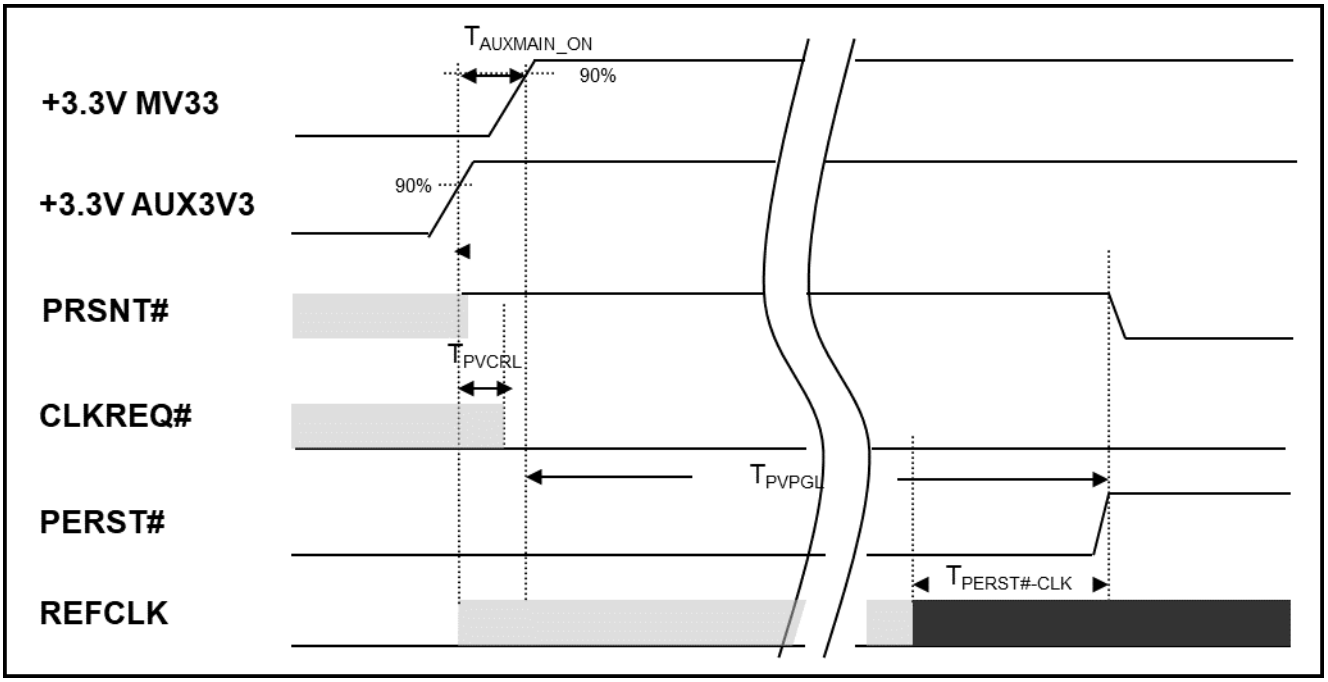


Figure 3. Chip Power-On or Resume from S3/S4

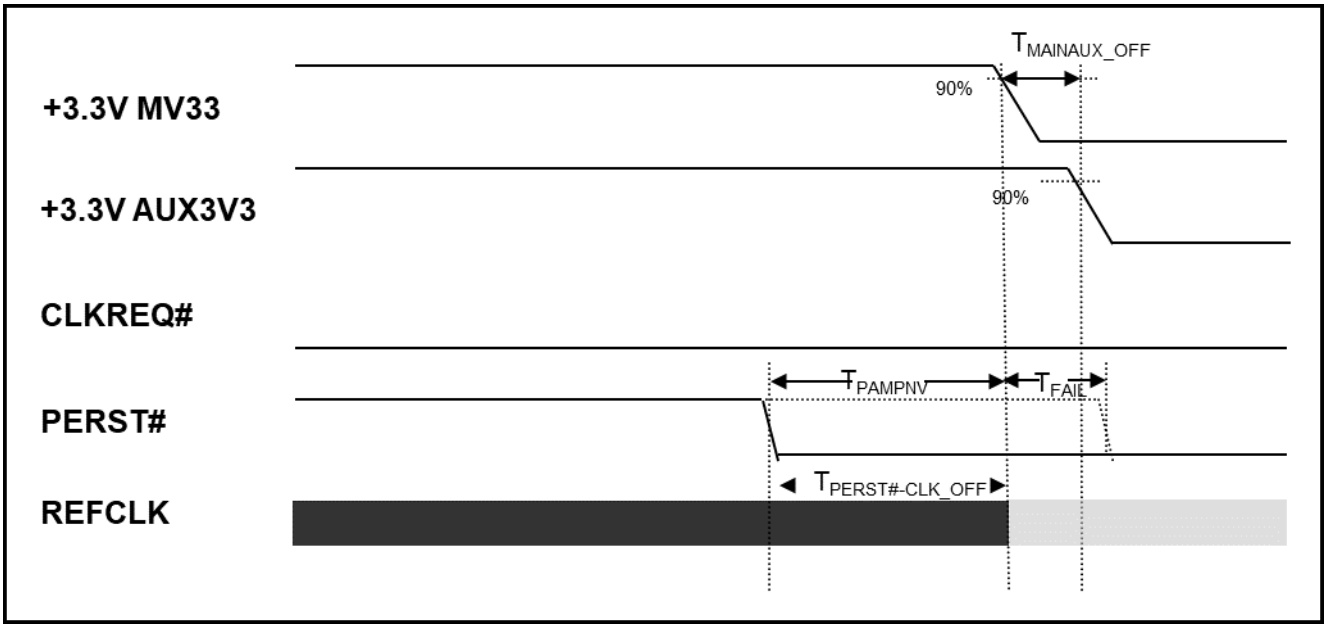


Figure 4. Chip Power-Off or Enter S3/S4

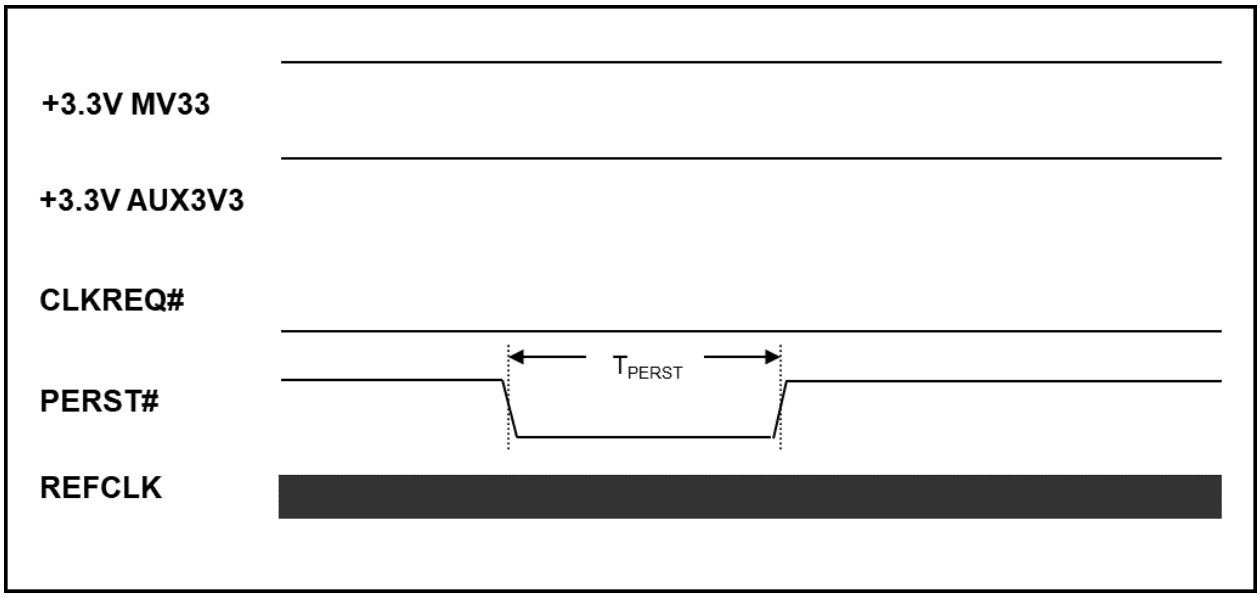


Figure 5. Chip Reboot

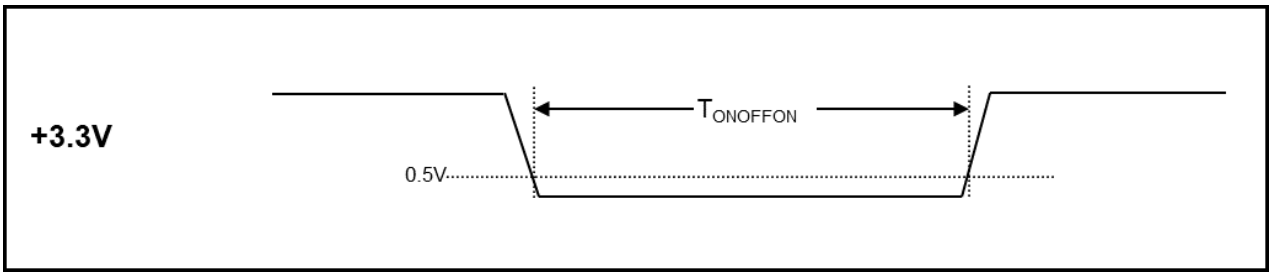


Figure 6. Chip Power Off then On

Table 85. Chip Power-On or Resume from S3/S4

Symbol	Description	Min.	Typ.	Max.	Unit
$T_{AUXMAIN_ON}$	Power-on timing between AUX3V3 to MV33 *MV33 power should not power on earlier than AUX3V3 power. Note that MV33 should be ready before memory card initialization. Refer to T_{PVPGL}.	0	-	-	ms
T_{PVCRL}	Power Valid to CLKREQ# output active *CLKREQ# might be low at the beginning.	-	-	100	µs
T_{PVPGL}	Power Valid to PERST# input inactive	1	-	-	ms
$T_{PERST\#-CLK}$	REFCLK stable before PERST# input inactive	100	-	-	µs
T_{POWER_RISE}	Power rise time for AUX3V3 and MV33 (10% to 90%)	0.5	-	15	ms

Table 86. Chip Power Off or enter S3/S4 Timing

Symbol	Description	Min.	Typ.	Max.	Unit
$T_{MAINAUX_OFF}$	Power-off timing between MV33 to AUX3V3 *AUX3V3 aux power should not power off earlier than MV33 main power.	0	-	-	ms
T_{FAIL}	Power level invalid to PERST# active *Power surprise remove	-	-	500	ns
T_{PAMPNV}	PERST# active to main power not valid	0	-	-	µs
T_{POWER_FALL}	Power fall time for AUX3V3 and MV33 (90% to 10%)	-	-	100	ms
$T_{PERST\#-CLK_OFF}$	PERST# active to REFCLK off	0	-	-	µs

Table 87. Chip Reboot

Symbol	Description	Min.	Typ.	Max.	Unit
T_{PERST}	PERST# active to PERST# inactive	100	-	-	µs

Table 88. Chip Power Off then On

Symbol	Description	Min.	Typ.	Max.	Unit
$T_{ONOFFON}$	Chip power OFF then ON *3.3V Power should be lower than 0.5V	2	-	-	ms

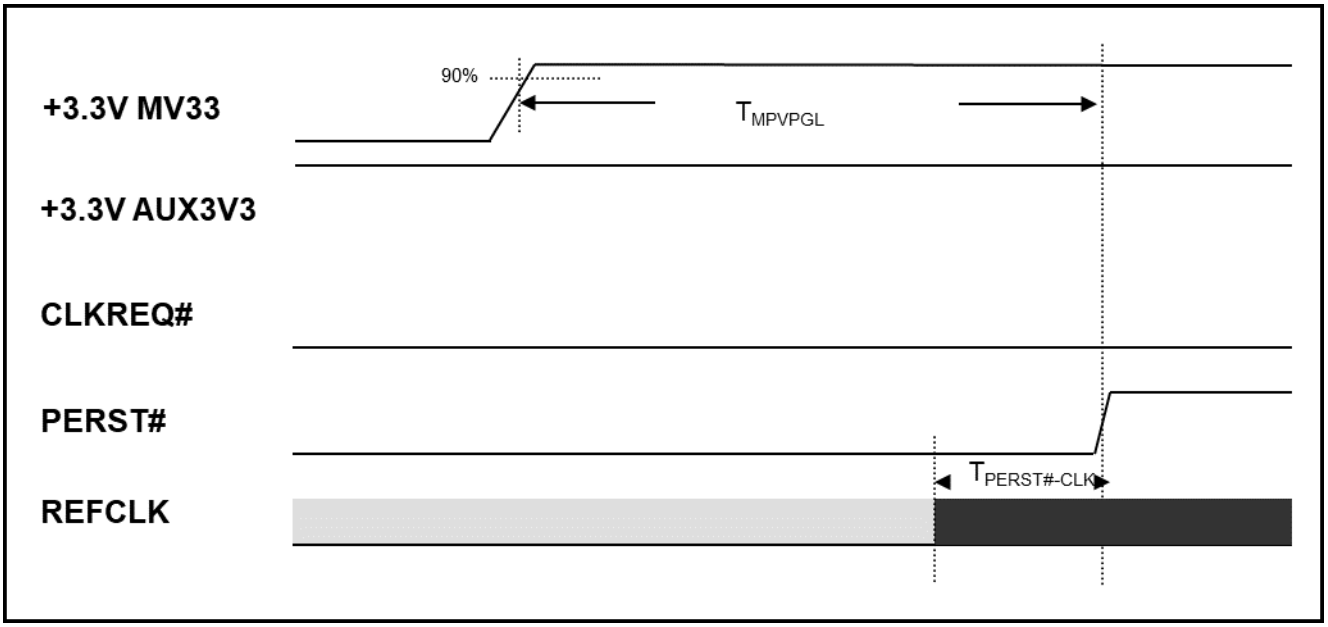


Figure 7. Chip RTD3 Resume

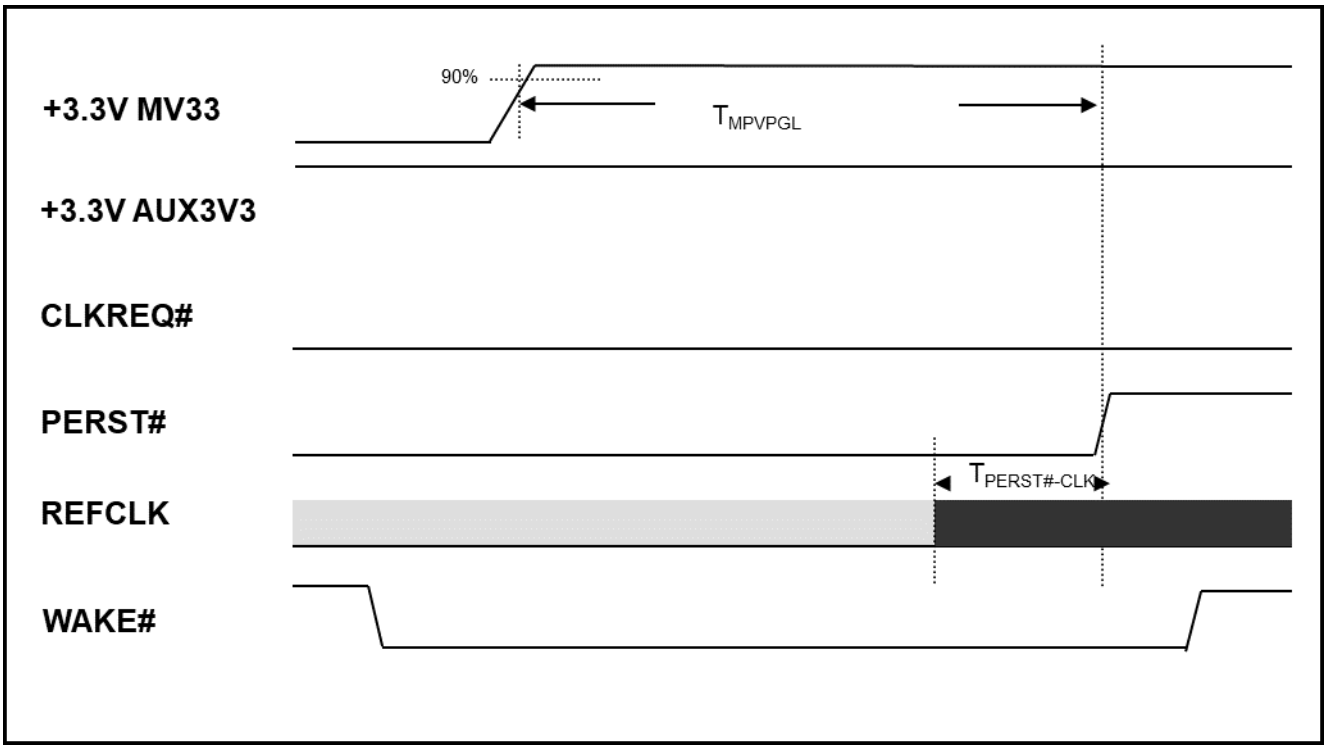


Figure 8. Chip RTD3 Resume by Wake

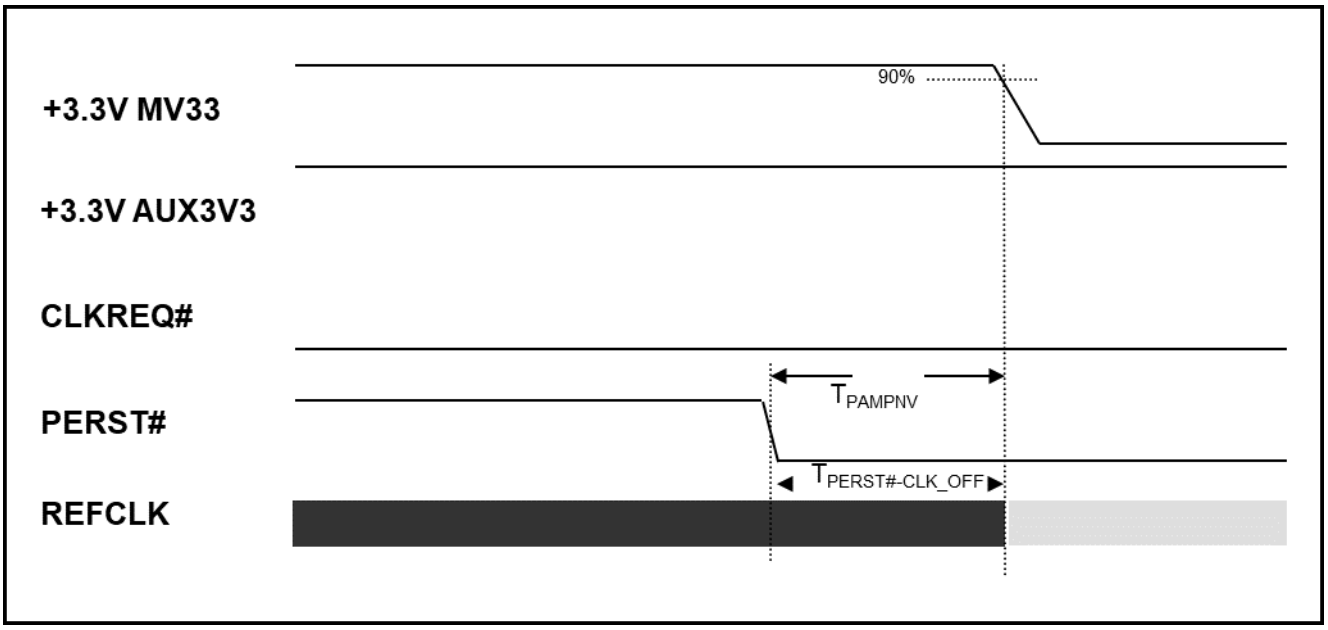


Figure 9. Chip RTD3 Enter

Table 89. Chip RTD3 Resume

Symbol	Description	Min.	Typ.	Max.	Unit
T_{MPVPGL}	MV33 Valid to PERST# input inactive	1	-	-	ms
T_{POWER_RISE}	Power rise time for AUX3V3 and MV33 (10% to 90%)	0.5	-	15	ms
$T_{PERST\#-CLK}$	REFCLK stable before PERST# input inactive	100	-	-	µs

Table 90. Chip RTD3 Resume by Wake

Symbol	Description	Min.	Typ.	Max.	Unit
T_{MPVPGL}	MV33 Valid to PERST# input inactive	1	-	-	ms
T_{POWER_RISE}	Power rise time for AUX3V3 and MV33 (10% to 90%)	0.5	-	15	ms
$T_{PERST\#-CLK}$	REFCLK stable before PERST# input inactive	100	-	-	µs

Table 91. Chip RTD3 Enter

Symbol	Description	Min.	Typ.	Max.	Unit
T_{PAMPNV}	PERST# active to main power not valid	0	-	-	µs
T_{POWER_FALL}	Power fall time for AUX3V3 and MV33	-	-	100	ms
$T_{PERST\#-CLK_OFF}$	PERST# active to REFCLK off	0	-	-	µs

10.3.2. Card Interface AC Characteristics

10.3.2.1 SD Interface - Default Speed Mode

Table 92. SD Interface - Default Speed Mode

Symbol	Description	Spec. (ns)	
		Min	Max
f_{PP}	Clock Frequency (Data Transfer Mode)	0	25MHz
f_{OD}	Clock Frequency (Identification Mode)	0	400KHz
t_{WL}	Clock Low Time	10	-
t_{WH}	Clock High Time	10	-
t_{TLH}	Clock Rise Time	-	10
t_{THL}	Clock Fall Time	-	10
t_{ISU}	Inputs CMD Setup Time	5	-
t_{IH}	Inputs CMD Hold Time	5	-
t_{ISU}	Inputs DATA Setup Time	5	-
t_{IH}	Inputs DATA Hold Time	5	-
t_{ODLY}	CMD Outputs Delay Time	0	14
t_{ODLY}	DATA Outputs Delay Time	0	14

10.3.2.2 SD Interface - High Speed Mode

Table 93. SD Interface - High Speed Mode

Symbol	Description	Spec. (ns)	
		Min	Max
f_{PP}	Clock Frequency (Data Transfer Mode)	0	50MHz
t_{WL}	Clock Low Time	7	-
t_{WH}	Clock High Time	7	-
t_{TLH}	Clock Rise Time	-	3
t_{THL}	Clock Fall Time	-	3
t_{ISU}	Inputs CMD Setup Time	6	-
t_{IH}	Inputs CMD Hold Time	2	-
t_{ISU}	Inputs DATA Setup Time	6	-
t_{IH}	Inputs DATA Hold Time	2	-
t_{ODLY}	CMD Outputs Delay Time	-	14
t_{OH}	CMD Outputs Hold Time	2.5	-
t_{ODLY}	DATA Outputs Delay Time	-	14
t_{OH}	DATA Outputs Hold Time	2.5	-

10.3.2.3 SD Interface - UHS-I DDR-50 Mode

Table 94. SD Interface - UHS-I DDR-50 Mode

Symbol	Description	Spec. (ns)	
		Min	Max
tCLK	Clock Period	20ns	-
tCR	Clock Rise Time	-	4.00ns
tCF	Clock Fall Time	-	4.00ns
tIS_DAT_r	Input DAT Setup Time rising edge	3.0ns	-
tIH_DAT_r	Input DAT Hold Time rising edge	0.8ns	-
tIS_DAT_f	Input DAT Setup Time falling edge	3.0ns	-
tIH_DAT_f	Input DAT Hold Time falling edge	0.8ns	-
tIS_CMD	Input CMD Setup Time	6.0ns	-
tIH_CMD	Input CMD Hold Time	0.8ns	-
tODLY_DAT_r	DAT Output Delay rising edge	-	7.0ns
tOH_DAT_r	Output DAT Hold time rising edge	1.5ns	-
tODLY_DAT_f	DAT Output Delay falling edge	-	7.0ns
tOH_DAT_f	Output DAT Hold time falling edge	1.5ns	-
tODLY_CMD	CMD Output Delay	-	13.7ns
tOH_CMD	Output CMD Hold time	1.5ns	-
-	Clock Duty	45%	55%

10.3.2.4 SD Interface - UHS-I SDR-50 Mode

Table 95. SD Interface - UHS-I SDR-50 Mode

Symbol	Description	Spec. (ns)	
		Min	Max
tCLK	Clock Period	4.8ns	-
tCR	Clock Rise Time	-	2.00ns
tCF	Clock Fall Time	-	2.00ns
tIS	Input DAT Setup Time	3.0ns	-
tIH	Input DAT Hold Time	0.8ns	-
tIS	Input CMD Setup Time	3.0ns	-
tIH	Input CMD Hold Time	0.8ns	-
tODLY	DAT Output Delay	-	7.5ns
tOH	Output DAT Hold time	1.5ns	-
tODLY	CMD Output Delay	-	7.5ns
tOH	Output CMD Hold time	1.5ns	-
-	Clock Duty	30%	70%

10.3.2.5 SD Interface - UHS-I SDR-104 Mode

Table 96. SD Interface - UHS-I SDR-104 Mode

Symbol	Description	Spec. (ns)	
		Min	Max
tCLK	Clock Period	4.8ns	-
tCR	Clock Rise Time	-	0.96ns
tCF	Clock Fall Time	-	0.96ns
tIS	Input DAT Setup Time	1.4ns	-
tIH	Input DAT Hold Time	0.8ns	-
tIS	Input CMD Setup Time	1.4ns	-
tIH	Input CMD Hold Time	0.8ns	-
tOP	DAT Output Phase	0	9.6ns
tODW	Output DAT Window	2.88ns	-
tOP	CMD Output Phase	0	9.6ns
tODW	Output CMD Window	2.88ns	-
-	Clock Duty	30%	70%

10.3.2.6 SD Interface - DDR 200MHz Mode

Table 97. DDR 200MHz Mode

Symbol	Description	Spec. (ns)	
		Min	Max
tCLK	Clock Period	5	-
tCR	Clock Rise Time	-	1
tCF	Clock Fall Time	-	1
tIS_DAT_r	Input DATA Setup Time rising edge	0.4	-

Symbol	Description	Spec. (ns)	
		Min	Max
tIH_DAT_r	Input DATA Hold Time rising edge	0.4	-
tIS_DAT_f	Input DATA Setup Time falling edge	0.4	-
tIH_DAT_f	Input DATA Hold Time falling edge	0.4	-
tIS_CMD	Input CMD Setup Time	0.4	-
tIH_CMD	Input CMD Hold Time	0.4	-
tODLY_DAT_r	DATA Output Delay rising edge	-	10
tODLY_DAT_f	DATA Output Delay falling edge	-	10
tODW	Output Data Window	1.2	-
tODLY_CMD	CMD Output Delay	-	10
tODW	Output CMD Window	1.2	
-	Clock Duty	44%	56%

10.3.2.7 SD Interface - DDR 225MHz Mode

Table 98. DDR 225MHz Mode

Symbol	Description	Spec. (ns)	
		Min	Max
tCLK	Clock Period	4.44	-
tCR	Clock Rise Time	-	0.888
tCF	Clock Fall Time	-	0.888
tIS_DAT_r	Input DATA Setup Time rising edge	0.4	-
tIH_DAT_r	Input DATA Hold Time rising edge	0.4	-
tIS_DAT_f	Input DATA Setup Time falling edge	0.4	-
tIH_DAT_f	Input DATA Hold Time falling edge	0.4	-
tIS_CMD	Input CMD Setup Time	0.4	-
tIH_CMD	Input CMD Hold Time	0.4	-
tODLY_DAT_r	DATA Output Delay rising edge	-	8.88
tODLY_DAT_f	DATA Output Delay falling edge	-	8.88
tODW	Output Data Window	1.0	-
tODLY_CMD	CMD Output Delay	-	8.88
tODW	Output CMD Window	1.0	
-	Clock Duty	44%	56%

10.3.2.8 SD Express Interface –PCIe Gen3 or PCIe Gen4 Mode

Refer to PCI-SIG specification: PCI Express® Base Specification Revision 4.0 and PCI Express® Card Electromechanical Specification Revision 4.0.

10.3.2.9 UHS-II Interface –FD156 and HD312

Refer to SD specification: Part 1 UHS-II Addendum Ver1.02.

10.3.3. High Speed MUX Electrical Characteristics

Table 99. High Speed MUX Electrical Characteristics

Symbol	Description	Min.	Typ.	Max.	Unit
Sdd21	MUX differential insertion loss at 4GHz	-	-1.9	-	dB
Sdd21	MUX differential insertion loss at 8GHz	-	-5.4	-	dB

10.4. Power Consumption

Table 100. Power Consumption

Host State	Device State	PCIe Link State	Reference Clock CLKREQ#	Power Consumption (mA)	Power Consumption (mW)	Note
S0	D0	L0	Asserted	136	449	UHS-I SDR104 R/W operation state w/o card's consumption
S0	D0	L0	Asserted	160	528	UHS-I DDR200 R/W operation state w/o card's consumption
S0	D0	L0	Asserted	235	776	UHS-II R/W operation state w/o card's consumption
S0	D0	L0 W/O ASPM	Asserted	116	383	Idle state w/o card
S0	D0	L0 W/O ASPM	Asserted	130	429	Idle state w UHS-I card
S0	D0	L0 W/O ASPM	Asserted	125	413	Idle state w UHS-II card
S0	D0	ASPM-L1.0	Asserted or De-asserted	0.9	2.97	Enter L1.0 state w/o card
S0	D0	ASPM-L1.0	Asserted or De-asserted	1	3.3	Enter L1.0 state w UHS-I card
S0	D0	ASPM-L1.0	Asserted or De-asserted	1.1	3.63	Enter L1.0 state w UHS-II card
S0	D0	ASPM-L1.1	De-asserted	0.4	1.32	Enter L1.1 state w/o card
S0	D0	ASPM-L1.1	De-asserted	0.65	2.15	Enter L1.1 state w UHS-I card
S0	D0	ASPM-L1.1	De-asserted	0.8	2.64	Enter L1.1 state w UHS-II card
S0	D0	ASPM-L1.2	De-asserted	0.25	0.83	Enter L1.2 state w/o card
S0	D0	ASPM-L1.2	De-asserted	0.55	1.82	Enter L1.2 state w UHS-I card
S0	D0	ASPM-L1.2	De-asserted	0.6	1.98	Enter L1.2 state w UHS-II card
S0/S0i3	D3	L1.2	De-asserted	0.25	0.83	Enter RTD3 hot mode w/o or w card
S0/S0i3	D3	L2	Asserted or De-asserted	0.07	0.23	Enter RTD3 cold mode w/o or w card
S3	D3	L1.0	Asserted or De-asserted	0.9	2.97	3.3V main&aux power keeps on w/o or w card
S3	D3	L1.1	De-asserted	0.4	1.32	3.3V main&aux power keeps on w/o or w card
S3	D3	L1.2	De-asserted	0.25	0.83	3.3V main&aux power keeps on w/o or w card

Host State	Device State	PCIe Link State	Reference Clock CLKREQ#	Power Consumption (mA)	Power Consumption (mW)	Note
S3	D3	L2	Asserted or De-asserted	0.07	0.23	3.3V main power off 3.3V AUX power on
W/O Card (Delink)			Asserted or De-asserted	0.07	0.23	Enter Auto-Delink mode
W/ SD Express card			Asserted or De-asserted	1.3	4.29	R/W or IDLE w/o card's consumption

Note 1: UHS-II or UHS-I Measurement at PCIe Gen II 5Gbps.
Note 2: With UHS-I SD DDR200/SDR104, UHS-II HD312 Card or SD Express Card.

10.5. Thermal Data

The user must select the correct thermal data according to your design.

For SD8.0 design, the thermal worse case is SD8.0 SD Express card.

Table 101. Thermal Data

Ta (°C)	Theta ja (°C/W)	Theta jc (°C/W)	Psi jt (°C/W)	PD(W)	Max Tj (°C)	Tc (°C)
70	37.8	19.1	0.39	1.18	114.6	92

Note 1: Including max power providing for SD8.0 SD Express card
Note 2: On 4-layer PCB

For SD7.x design, the thermal worse case is UHS-II card.

Table 102. Thermal Data

Ta (°C)	Theta ja (°C/W)	Theta jc (°C/W)	Psi jt (°C/W)	PD(W)	Max Tj (°C)	Tc (°C)
70	37.8	19.1	0.39	1.02	108.6	89.1

Note 1: Including max power providing for UHS-II card
Note 2: On 4-layer PCB

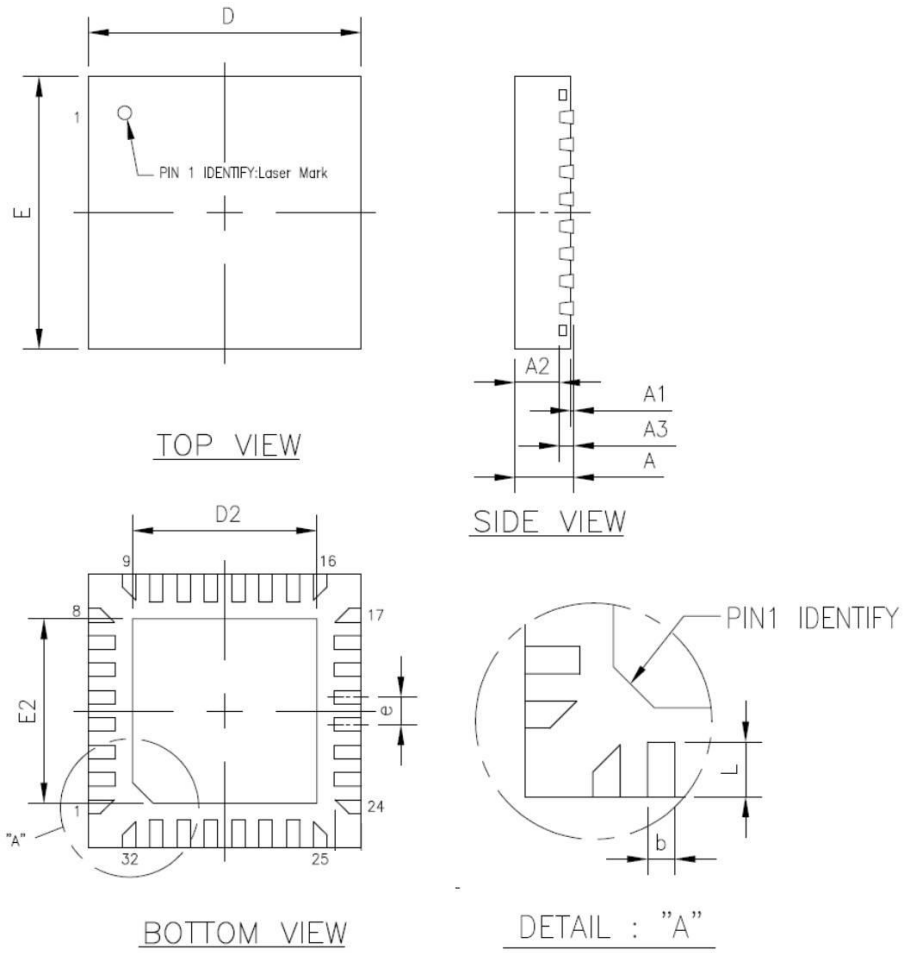
Table 103. Thermal Data

Ta (°C)	Theta ja (°C/W)	Theta jc (°C/W)	Psi jt (°C/W)	PD(W)	Max Tj (°C)	Tc (°C)
70	37.8	19.1	0.39	0.73	97.6	83.7

Note 1: Including max power providing for SD7.x SD Express card
Note 2: At 4-layer PCB

11. Mechanical Dimensions

Plastic Quad Flat No-Lead Package 32 Leads 4x4mm2 Outline



Symbol	Dimension in mm			Dimension in inch		
	Min	Nom	Max	Min	Nom	Max
A	0.80	0.85	0.90	0.031	0.033	0.035
A1	0.00	0.035	0.05	0.000	0.001	0.002
A2	—	0.65	0.70	—	0.026	0.028
A3	0.20 REF			0.008 REF		
b	0.15	0.20	0.25	0.006	0.0080	0.010
D/E	4.00 BSC			0.157 BSC		
D2/E2	2.55	2.70	2.85	0.096	0.106	0.116
e	0.40 BSC			0.016 BSC		
L	0.25	0.35	0.45	0.010	0.014	0.018

Note:

1. *CONTROLLING DIMENSION: MILLIMETER (mm).*
2. *REFERENCE DOCUMENT: JEDEC MO-220.*
3. *BSC/REF +/-0.1mm.*

12. Ordering Information

Part Number	Package	Status
RTS5264-GR	QFN32 32-pin Green package	Mass Production

Note: See page 6 for package identification.

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